# Eloralintide Fragment L2 Intermediate
# Global Process Flow Document (PFD)

LSN4182924, QA706D

This Process Flow Document was written, reviewed and approved as outlined below (reference SOP: APIEM-PRD-PFD Preparation, Approval and Maintenance (PRD-40523)):

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I have reviewed this Process Flow Document (PFD) and agree that the content and format satisfy SOP: APIEM-PRD-PFD Preparation, Approval and Maintenance.

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*Includes data integrity review which only includes changes from last revision

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## Revision History

| Version # | Version Date | Version Author | Reason for Revision |
|---|---|---|---|
| 1.0 | October 2026 | Johan Granander | Initial version. Includes PARs for Step 1 (SPPS) only. PARs for Step 2 (cleavage) are not yet complete and will be added in a future revision once defined. |
| 2.0 | October 2026 | Johan Granander | - Corrected the number of DMF washes after the Gly24 coupling and added the extended IPAc wash pre-Orn11 coupling to Table 1.2 and the PFC.<br>- Corrected MW for Fmoc-Phe-OH, Fmoc-Arg(Pbf)-OH and Fmoc-Leu-OH and the charge equivalents for NMeAsn22, Asn21 and αMePhe15 in Table 1.2.<br>- Corrected the LSN number for the specified endo-Gly24 impurity in Table 5.3.<br>- The coupling temperature target for Standard AAs, NMeAsn22 and αMePhe15 have been adjusted to 23 °C in the updated PAR report (version 2.0), which has been reflected in the PFC. |

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## References

The following documents were referenced to prepare this process flow document.

| Ref. # | Document Title | Version # | Document #/Ref |
|---|---|---|---|
| Ref. 1 | *Process Flow Document Preparation, Approval and Maintenance*; API External Manufacturing Operations Quality System Standard Operating Procedure | 5.0 | PRD-40523 |
| Ref. 2 | *Eloralintide Fmoc-Fragment 2 SPPS Risk Assessment and PAR Report* | 2.0 | RPT-1194035 |
| Ref. 3 | *Specification Rationale for Regulatory Starting Materials for Use in Eloralintide Drug Substance (LY3841136)* | 1.0 | RPT-1196579 |
| Ref. 4 | *Specification Rationale for Fmoc-Fragment L2 LSN4182924, GMP Intermediate for LY3841136 Eloralintide* | 1.0 | RPT-1195540 |

## Abbreviations

| Abbreviation | Definition | Abbreviation | Definition |
|---|---|---|---|
| **2-CTC** | 2-Chlorotrityl Chloride (Resin) | **L** | Liter(s) |
| **AA** | Amino Acid | **L2** | Fragment 2 for Eloralintide |
| **Ac2O** | Acetic Anhydride | **LAARA** | Eloralintide |
| **ACN** | Acetonitrile | **LCMS** | Liquid Chromatography – Mass Spectrometry |
| **AFD** | Agitated Filter Dryer | **LDPE** | Low-Density Polyethylene |
| **Ala / A** | Alanine | **Leu / L** | Leucine |
| **APIEM** | Active Pharmaceutical Ingredient External Manufacturing | **LOD** | Loss on Drying |
| **API** | Active Pharmaceutical Ingredient | **LR** | Limiting Reagent |
| **AR** | Acceptable Range | **Lys / K** | Lysine |
| **Arg / R** | Arginine | **NMeAsn** | N-Methylated Asparagine |
| **Asn / N** | Asparagine | **MeOH** | Methanol |
| **Asp / D** | Aspartic Acid | **M.P.** | Melting Point |
| **B.P.** | Boiling Point | **M.W.** | Molecular Weight |
| **Boc** | t-Butyloxycarbonyl | **min** | Minute(s) |
| **BR&D** | Bioprocess Research & Development | **mmol** | Millimole(s) |
| **CIPC** | Critical In-Process Control | **mol** | Mole(s) |
| **CM** | Contract Manufacturer | **MOR** | Manufacturing Operating Range |
| **CoA** | Certificate of Analysis | **MS** | Mass Spectrometry/spectrometer |
| **CPP** | Critical Process Parameter | **MTBE** | Methyl tert-Butyl Ether |
| **CQA** | Critical Quality Attribute | **NaCl** | Sodium Chloride |
| **d** | Density | **NLT** | Not Less Than |
| **DBF** | Dibenzofulvene | **NMT** | Not More Than |
| **DCM** | Dichloromethane | **NOR** | Normal Operating Range |
| **des** | “without” – molecule missing referenced amino acid(s) or functional group(s) | **OEL** | Occupational Exposure Limit |
| **DIC** | 1,3-Diisopropylcarbodiimide | **Orn** | Ornithine |

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| | | | |
|---|---|---|---|
| **DIPEA / DIEA** | *N*,*N*-Diisopropylethylamine | **Oxyma** | Ethyl 2-cyano-2-(hydroxyamino)acetate |
| **DIU** | Diisopropylurea | **PAR** | Proven Acceptable Range |
| **DKP** | Diketopiperazine | **Pbf** | 2,2,4,6,7-Pentamethyldihydrobenzofuran-5-sulfonyl |
| **DMF** | *N*,*N*-Dimethylformamide | **PFC** | Process Flow Chart |
| **EFC** | Equipment Flow Chart | **PFD** | Process Flow Document |
| **eq.** | Molar Equivalent(s) | **Phe / F** | Phenylalanine |
| **EtOAc** | Ethyl Acetate | **PIP** | Piperidine |
| **EtOH** | Ethanol | **Pro / P** | Proline |
| **FIO** | For Information Only | **PyBOP** | Benzotriazol-1-yl-oxytripyrrolidinophosphonium Hexafluorophosphate |
| **Fmoc** | 9-Fluorenylmethyloxycarbonyl | **Pyr** | Pyridine |
| **g** | Gram(s) | **QA** | Quality Assurance |
| **GC** | Gas Chromatography | **ROE** | Residue on Evaporation |
| **GIP** | Glucose-dependent Insulinotropic Peptide | **rOPP** | Registered Operating Parameter |
| **GL** | Glass | **Ser / S** | Serine |
| **Gln / Q** | Glutamine | **SMDD** | Synthetic Molecule Design & Development |
| **GLP** | Glucagon-like Peptide | **SOP** | Standard Operating Procedure |
| **Glu / E** | Glutamic Acid | **SPPS** | Solid Phase Peptide Synthesis |
| **Gly / G** | Glycine | **SS** | Stainless Steel |
| **h** | Hour(s) | **tBu** | tert-Butyl |
| **HCN** | Hydrogen Cyanide | **TFA** | Trifluoroacetic Acid |
| **His / H** | Histidine | **THF** | Tetrahydrofuran |
| **HOBt** | Hydroxybenzotriazole | **Thr / T** | Threonine |
| **HPLC** | High-Performance Liquid Chromatography | **Trt** | Trityl |
| **IBC** | Individual Bulk Container | **TS/MS** | Technical Services / Manufacturing Science |
| **Ile / I** | Isoleucine | **Tyr / Y** | Tyrosine |
| **IPA** | Isopropanol | **UPLC** | Ultra-Performance Liquid Chromatography |
| **IPAc** | Isopropyl Acetate | **V** | Volume(s) (L/kg) |
| **IPC** | In-Process Control | **Val / V** | Valine |
| **KF** | Karl Fischer | **α-Me-Phe** | α-Methylphenylalanine |
| **Kg** | Kilogram(s) | | |

* Definitions for these terms can be found in the Lilly Quality System Glossary.

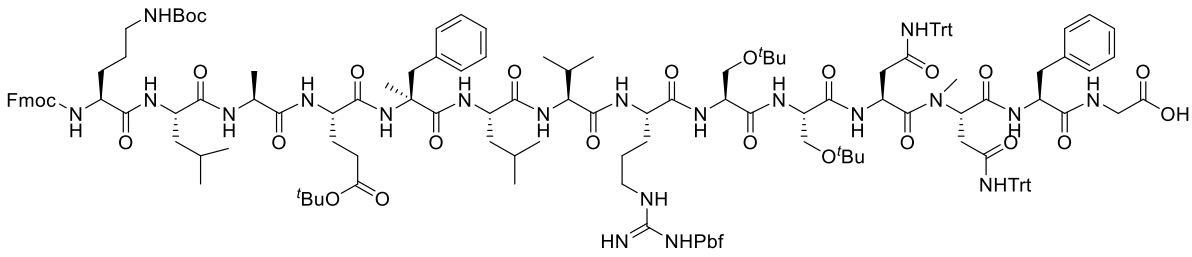
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# 1. Process Chemistry

## 1.1 Introduction

Eloralintide (LAARA) Fragment L2 intermediate (LSN4182924) is a synthetically derived peptide consisting of a fourteen-residue backbone, which includes a Trt-protected asparagine and N-methylated asparagine which is synthesized as an RSM (ASYM-138520) by Asymchem. It also contains tBu-protected glutamic acid and serine, Boc protected ornithine and Pbf protected arginine.

The Fragment L2 intermediate backbone is manufactured using solid phase peptide synthesis (SPPS), followed by cleavage from the resin and product isolation. The RSM, Fmoc-N-Me-Asn(Trt)-OH, is manufactured separately in a 3-step synthetic process.




Chemical Formula: $C_{155}H_{200}N_{20}O_{28}S$
Exact Mass: 2821.46
Molecular Weight: 2823.48

**Figure 1.1:** Eloralintide Fragment 2 Intermediate (LSN4182924)

LAARA Fragment L2 intermediate (L2) is used within a hybrid process sequence to synthesize Eloralintide drug substance (LY3841136), which is a long-acting peptide with selective, potent agonist activity on human amylin receptor 1.

## 1.2 Synthetic Route

Fragment L2 intermediate is prepared using solid phase peptide synthesis (SPPS) on 2-chlorotrityl chloride (CTC) resin. SPPS elongation begins at residue 24 and proceeds to residue 11. Before assembly, the resin is swelled in dimethylformamide (DMF) and loaded with Fmoc-Gly-OH under temperature-controlled conditions with nitrogen sparging to support efficient substitution. Unreacted sites on the resin are then capped using methanol and diisopropylethylamine (DIPEA), followed by DMF washing to prepare the resin for elongation.

Peptide chain growth proceeds through iterative SPPS cycles that include amino acid activation and coupling in DMF using the amino acid derivative, Oxyma, and 1,3-diisopropylcarbodiimide (DIC); verification of coupling completion using a ninhydrin-, chloranil- or bromophenol blue-based IPC test; washing the resin with DMF; removing the Fmoc protecting group using 20% piperidine/DMF; and washing again with isopropyl acetate (IPAc) and DMF. These cycles continue until the full Fragment L2 peptide sequence has been assembled.

Certain amino acids require modified activation or coupling conditions. For example, Fmoc-L-Phe-OH is coupled at reduced temperature using PyBOP/Oxyma to prevent side-reactions, and reactions time vary depending on amino acid complexity. In the case of Asn21, a capping solution of acetic anhydride and pyridine is applied to suppress formation of deletion sequences.

Throughout synthesis, reactor temperature, oxygen levels, and agitation performance are carefully monitored to maintain consistent peptide assembly.

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Once elongation is complete, the peptide-resin is washed using IPAc, toluene, and methyl tert-butyl ether (MTBE), then dried under vacuum and nitrogen sweep to achieve the required residual loss-on-drying target. The dried peptide-resin may be held under controlled conditions before cleavage.

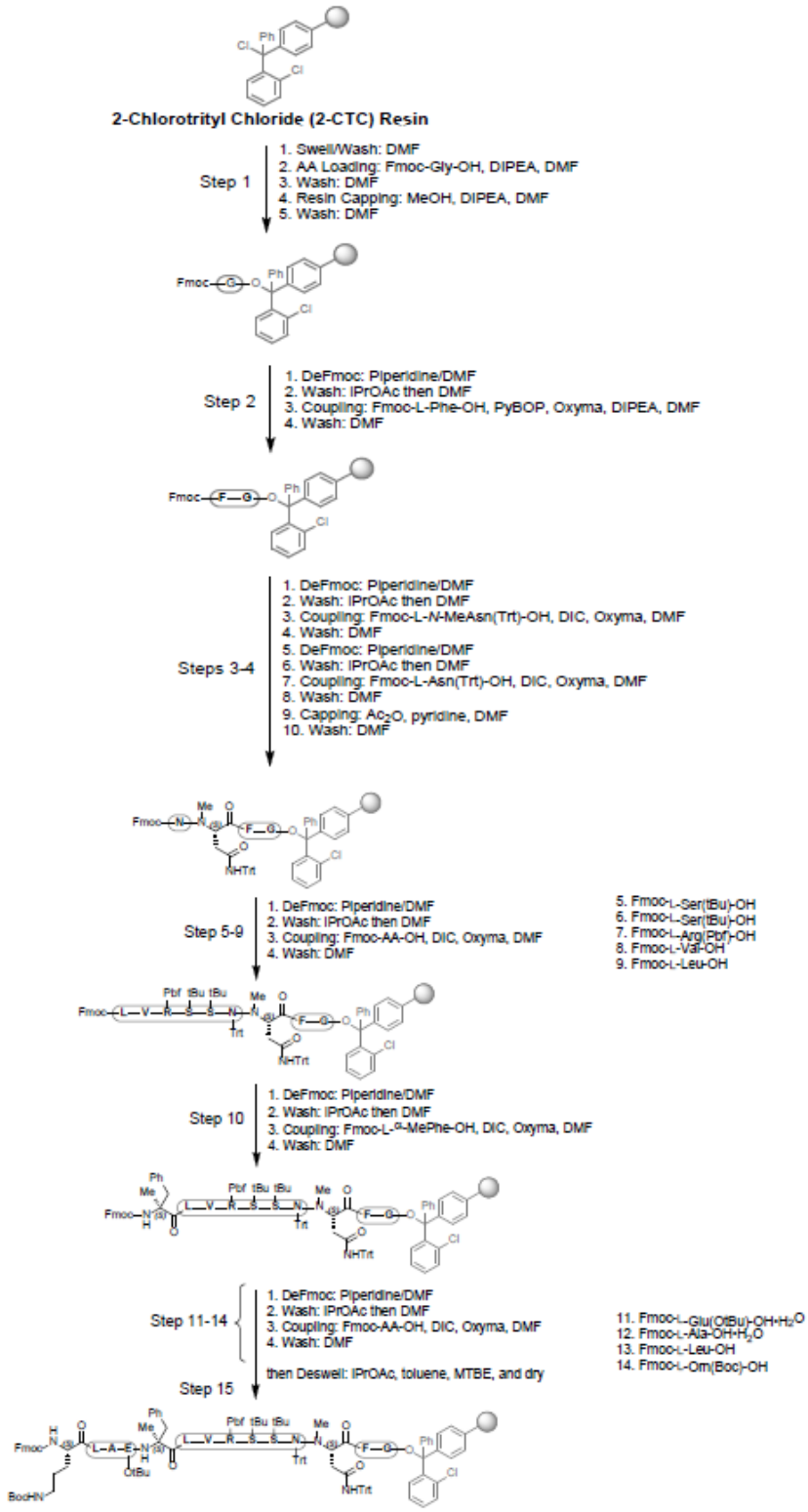



**Figure 1.2:** Synthetic route for producing Fragment L2 on resin

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**Table 1.2:** Materials and stoichiometry used in the SPPS synthesis of LAARA Fragment L2 on resin

| Material | M.W. (g/mol) | B.P. (°C) | M.P. (°C) | Density (kg/L) | Step/Cycle | Eq. (mol/mol LR*) | w/w (kg/kg LR*) |
|---|---|---|---|---|---|---|---|
| **Resin swell** | | | | | | | |
| 2-CTC resin | N/A | N/A | N/A | N/A | Resin loading | 1.00 | 1.00 |
| *N*,*N*-Dimethylformamide (DMF) | 73.09 | 153 | -61 | 0.95 | Pre-treatment | N/A | 5.21 |
| | | | | | Resin swell | | 7.58 |
| | | | | | Post swell wash | | 4.74 |
| **Gly24 activation** | | | | | | | |
| Fmoc-Gly-OH | 297.31 | N/A | 174-175 | N/A | Gly24 activation | 2.25 | N/A |
| *N*,*N*-Dimethylformamide (DMF) | 73.09 | 153 | -61 | 0.95 | Gly24 activation | N/A | 4.98 |
| *N*,*N*-Diisopropylethylamine (DIPEA) | 129.24 | 127 | -50 | 0.74 | Gly24 activation | 6.00 | N/A |
| **Gly24 coupling** | | | | | | | |
| *N*,*N*-Dimethylformamide (DMF) | 73.09 | 153 | -61 | 0.95 | Gly24 coupling | N/A | 5.93 |
| | | | | | Gly24 post coupling wash | | 7.58 |
| **Gly24 capping** | | | | | | | |
| *N*,*N*-Dimethylformamide (DMF) | 73.09 | 153 | -61 | 0.95 | Gly24 capping | N/A | 8.53 |
| | | | | | Post capping wash | | 3× 7.58 |
| Methanol | 32.04 | 65 | -98 | 0.79 | Gly24 capping | 10.00 | N/A |
| *N*,*N*-Diisopropylethylamine (DIPEA) | 129.24 | 127 | -50 | 0.74 | Gly24 capping | 5.00 | |
| **Deprotection** | | | | | | | |
| 20% Piperidine in DMF | 85.15** | N/A | N/A | 0.93 | Standard cycle | N/A | 7.45 |
| | | | | | Pre-Ser20 Treatment 1 | | 7.45 |
| | | | | | Pre-Ser20 Treatment 2 | | 9.68 |
| Isopropyl acetate | 102.13 | 85-91 | −73 | 0.87 | Post deprotection wash (Standard cycle) | | 4× NLT 8.73 |
| | | | | | Post deprotection wash (Pre-Orn11 coupling) | | 10× NLT 8.73 |
| *N*,*N*-Dimethylformamide (DMF) | 73.09 | 153 | -61 | 0.95 | Re-swelling (Standard cycle) | | 2× NLT 11.38 |
| | | | | | Re-swelling (Pre-Ser20 coupling) | | 2× NLT 12.32 |
| | | | | | Re-swelling (Pre-Val17, Pre-Leu12 coupling) | | 2× NLT 9.48 |
| **Activation** | | | | | | | |
| *N*,*N*-Dimethylformamide (DMF) | 73.09 | 153 | -61 | 0.95 | Standard cycle activation | N/A | 6.64 |
| | | | | | Phe23 activation | | 4.74 |
| | | | | | NMeAsn22, αMePhe15 activation | | 5.69 |
| | | | | | Asn21 | | 5.69 |
| Fmoc-L-Phe-OH | 387.43 | N/A | 180-187 | N/A | Phe23 activation | 2.00 | N/A |
| Fmoc-L-N-Me-Asn(Trt)-OH | 610.70 | | | | NMeAsn22 activation | 1.75 | |
| Fmoc-L-Asn(Trt)-OH | 596.67 | | 201-204 | | Asn21 activation | 2.50 | |
| Fmoc-L-Ser(tBu)-OH | 383.44 | | | | Ser20 activation | 2.00 | |
| | | | | | Ser19 activation | 2.00 | |
| Fmoc-L-Arg(Pbf)-OH | 648.77 | | | | Arg18 activation | 2.00 | |
| Fmoc-L-Val-OH | 339.39 | | 143-147 | | Val17 activation | 2.00 | |
| Fmoc-L-Leu-OH | 353.41 | | 152-156 | | Leu16 activation | 2.00 | |

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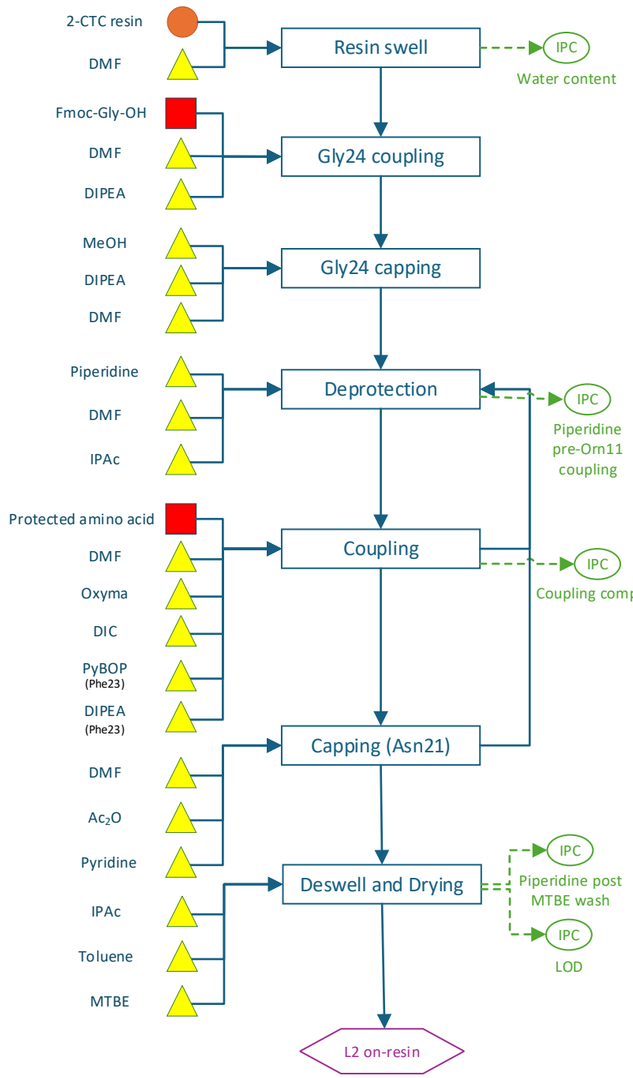
| Material | M.W. (g/mol) | B.P. (°C) | M.P. (°C) | Density (kg/L) | Step/Cycle | Eq. (mol/mol LR*) | w/w (kg/kg LR*) |
|---|---|---|---|---|---|---|---|
| | | | | | Leu12 activation | 2.00 | |
| Fmoc-L-α-Me-Phe-OH | 401.45 | | | | αMePhe15 activation | 1.75 | |
| Fmoc-L-Glu(OtBu)-OH·$H_2O$ | 443.49 | | 108-112 | | Glu14 activation | 2.00 | |
| Fmoc-L-Ala-OH·$H_2O$ | 329.35 | | 155-158 | | Ala13 activation | 2.00 | |
| Fmoc-L-Orn(Boc)-OH | 454.52 | | | | Orn11 activation | 2.00 | |
| Ethyl cyano(hydroxyimino)acetate (Oxyma) | 142.11 | | 130-132 | | Standard cycle activation | 2.00 | |
| | | | | | Phe23 activation | 2.00 | |
| | | | | | NMeAsn22, αMePhe15 activation | 1.80 | |
| | | | | | Asn21 activation | 2.60 | |
| PyBOP | 520.39 | | 154-156 | | Phe23 activation | 2.00 | |
| *N*,*N*-Diisopropylethylamine (DIPEA) | 129.24 | 127 | -50 | 0.74 | Phe23 activation | 3.00 | |
| 1,3-Diisopropylcarbodiimide (DIC) | 126.20 | 145-148 | N/A | 0.81 | Standard cycle activation | 2.20 | |
| | | | | | NMeAsn22, αMePhe15 activation | 2.00 | |
| | | | | | Asn21 activation | 2.90 | |
| **Coupling** | | | | | | | |
| *N*,*N*-Dimethylformamide (DMF) | 73.09 | 153 | -61 | 0.95 | Total DMF (Standard cycle) | N/A | 7.58 |
| | | | | | Total DMF (Phe23) | | 7.58 |
| | | | | | Total DMF (NMeAsn22, αMePhe15) | | 6.64 |
| | | | | | Total DMF (Asn21) | | 7.58 |
| | | | | | Post-coupling wash (except Orn11) | | 3× 7.58 |
| Ethyl cyano(hydroxyimino)acetate (Oxyma) | 142.11 | N7A | 130-132 | N/A | Kicker charge Asn21 | 1.00 | N/A |
| 1,3-Diisopropylcarbodiimide (DIC) | 126.20 | 145-148 | N/A | 0.81 | Kicker charge Asn21 | 1.10 | |
| **Capping post Asn21 coupling** | | | | | | | |
| *N*,*N*-Dimethylformamide (DMF) | 73.09 | 153 | -61 | 0.95 | Asn21 capping | N/A | 6.64 |
| | | | | | Post capping wash | | 3 × 7.58 |
| Acetic anhydride | 102.09 | 139 | -73 | 1.08 | Asn21 capping | 1.25 | N/A |
| Pyridine | 79.10 | 115 | -42 | 0.98 | Asn21 capping | 1.50 | |
| **Deswell** | | | | | | | |
| Isopropyl acetate (IPAc) | 102.13 | 89 | -73 | 0.87 | IPAc deswell | N/A | 5x NLT 8.73 |
| Toluene | 92.14 | 110.6 | -95 | 0.87 | Toluene deswell | | 2x NLT 8.72 |
| Methyl tert-Butyl Ether (MTBE) | 88.15 | 55-56 | -110 | 0.74 | MTBE deswell | | 4x NLT 7.41 |

*Limiting reagent.
**M.W. for piperidine

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## 2. Process Flow Chart (PFC)




| ELORALINTIDE FRAGMENT L2 INTERMEDIATE | CONTROL PARAMETER | TARGET | PAR |
|---|---|---|---|
| | **RESIN SWELL** | | |
| | Pre-treatment DMF quantity | 5.21 kg/kg | N/A |
| | Water content (IPC) | NMT 500 ppm | N/A |
| | 2-CTC resin substitution degree | 1.0 – 1.2 mmol/g | N/A |
| | DMF swelling quantity | 7.58 kg/kg | N/A |
| | Swelling temperature | 25 °C | N/A |
| | Swelling time | 35 min | N/A |
| | DMF wash quantity | 4.74 kg/kg | N/A |
| | **RESIN LOADING and CAPPING** | | |
| | Activation – Gly24 quantity | 2.25 eq. | 2.12 – 2.54 eq. |
| | Activation – DMF quantity | 4.98 kg/kg | 4.14 – 5.61 kg/kg |
| | **Activation – DIPEA quantity** | **6.00 eq.** | 5.24 – **6.69 eq.** |
| | **Activation – Dissolution temperature** | **25.0 °C** | **17.0 – 33.8 °C** |
| | Activation – Dissolution time | 5 min | 0 – 66 min |
| | Coupling – Total DMF quantity | 5.93 kg/kg | 4.97 – 6.68 kg/kg |
| | **Coupling – Reaction temperature** | **25.0 °C** | **17.2 – 32.7 °C** |
| | Coupling – Reaction time | NLT 180 min | 180 – 481 min |
| | Coupling – DMF wash quantity | NLT 7.58 kg/kg | NLT 6.62 kg/kg |
| | Capping – Total DMF quantity | 8.53 kg/kg | 7.47 – 9.61 kg/kg |
| | Capping – MeOH quantity | 10.00 eq. | 8.73 – 11.20 eq. |
| | Capping – DIPEA quantity | 5.00 eq. | 4.36 – 5.60 eq. |
| | Capping – Temperature | 25.0 °C | 13.0 – 36.8 °C |
| | Capping – Time | 60 min | 15 – 122 min |
| | **Capping – DMF wash quantity** | **3x NLT 7.58 kg/kg** | **3x NLT 7.58 kg/kg** |
| | **DEPROTECTION** | | |
| | Piperidine/DMF solution concentration | 20% v/v | 14.99 – 25.00% v/v |
| | Piperidine/DMF solution quantity (Standard AA) | 7.45 kg/kg | 6.54 – 8.38 kg/kg |
| | Piperidine/DMF solution quantity (Pre-Ser20 Treatment 1) | 7.45 kg/kg | 6.51 – 8.40 kg/kg |
| | Piperidine/DMF solution quantity (Pre-Ser20 Treatment 2) | 9.68 kg/kg | 7.44 – 10.91 kg/kg |
| | **Deprotection temperature (Standard AA)** | **25.0 °C** | **17.0 – 34.8 °C** |
| | **Deprotection temperature (Pre-Ser20)** | **25.0 °C** | **14.6 – 34.7 °C** |
| | Piperidine/DMF exposure time (Standard AA) | 120 min | 68 – 180 min |
| | **Piperidine/DMF exposure time (Pre-Ser20 Treatment 1)** | **15 min** | **6 – 68 min** |
| | **Piperidine/DMF exposure time (Pre-Ser20 Treatment 2)** | **30 min** | **6 – 83 min** |
| | **IPAc wash quantity (Standard AA)** | **4x NLT 8.73 kg/kg** | **4x NLT 7.93 kg/kg** |
| | **IPAc wash quantity (Pre-Orn11 coupling)** | **10x NLT 8.73 kg/kg** | **10x NLT 7.93 kg/kg** |
| | Chloranil Test (IPC) (Pre-Orn11 coupling) | Negative | N/A |
| | DMF wash quantity (Standard AA) | 2x NLT 11.38 kg/kg | 2x NLT 9.96 kg/kg |
| | DMF wash quantity (Pre-Ser20 coupling) | 2x NLT 12.32 kg/kg | 2x NLT 10.78 kg/kg |
| | DMF wash quantity (Pre-Val17, Pre-Leu12 coupling) | 2x NLT 9.48 kg/kg | 2x NLT 8.29 kg/kg |
| | **ACTIVATION** | | |
| | AA quantity (Standard AA) | 2.00 eq. | 1.88 – 2.13 eq. |
| | AA quantity (Phe23) | 2.00 eq. | 1.88 – 2.13 eq. |
| | AA quantity (NMeAsn22, αMePhe15) | 1.75 eq. | 1.65 – 1.86 eq. |

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| | | |
|---|---|---|
| AA quantity (Asn21) | 2.50 eq. | 2.35 – 2.66 eq. |
| **Oxyma quantity (Standard AA)** | **2.00 eq.** | **1.94** – 2.25 **eq.** |
| Oxyma quantity (Phe23) | 2.00 eq. | 1.88 – 2.26 eq. |
| **Oxyma quantity (NMeAsn22, αMePhe15)** | **1.80 eq.** | **1.75** – 2.03 **eq.** |
| **Oxyma quantity (Asn21)** | **2.60 eq.** | **2.50** – 2.93 **eq.** |
| **Dissolution temperature (Standard AA)** | **20.0 °C** | **12.6 – 26.5 °C** |
| **Dissolution temperature (Phe23)** | **20.0 °C** | **12.0 – 29.5 °C** |
| **Dissolution temperature (NMeAsn22, αMePhe15)** | **20.0 °C** | **12.6 – 26.9 °C** |
| **Dissolution temperature (Asn21)** | **20.0 °C** | **12.2 – 27.9 °C** |
| PyBOP quantity (Phe23 cycle) | 2.00 eq. | 1.87 – 2.25 eq. |
| DIPEA quantity (Phe23 cycle) | 3.00 eq. | 2.62 – 3.40 eq. |
| **DIC quantity (Standard AA)** | **2.20 eq.** | **2.01** – 2.48 **eq.** |
| DIC quantity (NMeAsn22, αMePhe15) | 2.00 eq. | 1.75 – 2.27 eq. |
| DIC quantity (Asn21) | 2.90 eq. | 2.52 – 3.29 eq. |
| Total activation DMF quantity (Standard AA) | 6.64 kg/kg | 5.84 – 7.46 kg/kg |
| Total activation DMF quantity (Phe23) | 4.74 kg/kg | 4.16 – 5.33 kg/kg |
| Total activation DMF quantity (NMeAsn22, αMePhe15) | 5.69 kg/kg | 5.00 – 6.41 kg/kg |
| Total activation DMF quantity (Asn21) | 5.69 kg/kg | 4.98 – 6.40 kg/kg |
| **Activation temperature (Standard AA)** | **20.0 °C** | **12.6 – 26.5 °C** |
| **Activation temperature (Phe23)** | **5.0 °C** | **-2.5 – 13.6 °C** |
| **Activation temperature (NMeAsn22, αMePhe15)** | **20.0 °C** | **12.6 – 26.9 °C** |
| **Activation temperature (Asn21)** | **20.0 °C** | **12.2 – 27.9 °C** |
| Activation time (Standard AA) | 20 min | 0 – 80 min |
| Activation time (Phe23) | 20 min | 0 – 80 min |
| Activation time (NMeAsn22, αMePhe15) | 20 min | 0 – 80 min |
| Activation time (Asn21) | 20 min | 0 – 80 min |
| **COUPLING** | | |
| Total coupling DMF quantity (Standard AA) | 7.58 kg/kg | 6.67 – 8.53 kg/kg |
| Total coupling DMF quantity (Phe23) | 7.58 kg/kg | 4.99 – 8.54 kg/kg |
| Total coupling DMF quantity (NMeAsn22, αMePhe15) | 6.64 kg/kg | 5.83 – 7.49 kg/kg |
| Total coupling DMF quantity (Asn21) | 7.58 kg/kg | 6.65 – 8.55 kg/kg |
| Oxyma kicker charge quantity (Asn21 cycle) | 1.00 eq. | 0.94 – 1.13 eq. |
| DIC kicker charge quantity (Asn21 cycle) | 1.10 eq. | 0.96 – 1.26 eq. |
| **Coupling temperature (Standard AA)** | **23.0 °C** | **18.6 – 28.4 °C** |
| **Coupling temperature (Phe23)** | **25.0 °C** | **17.4 – 30.3 °C** |
| **Coupling temperature (NMeAsn22, αMePhe15)** | **23.0 °C** | **17.3 – 28.3 °C** |
| **Coupling temperature (Asn21)** | **25.0 °C** | **17.2 – 30.2 °C** |
| Coupling reaction time (Standard AA) | NLT 240 min | 240 – 600 min |
| Coupling reaction time (Phe23) | NLT 360 min | 360 – 841 min |
| Coupling reaction time (NMeAsn22, αMePhe15) | NLT 360 min | 360 – 840 min |
| Total Coupling reaction time (Asn21) | NLT 780 min | 780 – 1680 min |
| Coupling reaction time (Glu14) | NLT 480 min | 480 – 1083 min |
| Colorimetric Test (IPC) (Ninhydrin for Phe23, NMeAsn22, Ser20-αMePhe15, Ala13-Orn11; Chloranil for Asn21; Bromophenol Blue for Glu14) | Negative (x2) | N/A |
| Post coupling DMF wash quantity (except Orn11) | 3x NLT 7.58 kg/kg | 3x NLT 6.62 kg/kg |
| **CAPPING POST ASN21 COUPLING** | | |
| Capping DMF quantity | 6.64 kg/kg | 5.80 – 7.47 kg/kg |

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| | | | |
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| | Capping Acetic anhydride quantity | 1.25 eq. | 1.09 – 1.41 eq. |
| | Capping Pyridine quantity | 1.50 eq. | 1.31 – 1.69 eq. |
| | Capping reaction temperature | 25.0 °C | 12.3 – 36.8 °C |
| | Capping reaction time | 60 min | 10 – 120 min |
| | **Post capping DMF wash quantity** | **3x NLT 7.58 kg/kg** | **3x NLT 6.75 kg/kg** |
| | **DE-SWELL AND DRYING** | | |
| | **IPAc wash quantity** | **5x NLT 8.73 kg/kg** | **5x NLT 8.73 kg/kg** |
| | **Toluene wash quantity** | **2x NLT 8.72 kg/kg** | **2x NLT 8.72 kg/kg** |
| | **MTBE wash quantity** | **4x NLT 7.41 kg/kg** | **4x NLT 7.41 kg/kg** |
| | Chloranil Test (IPC) | Negative | N/A |
| | Drying jacket temperature | NMT 30.0 °C | NMT 42.0 °C |
| | LOD IPC | NMT 9% w/w | N/A |


Non-Critical Raw Material


Critical Raw Material


Registered Starting Material


Product

( Sample )

**Highlighted: - Critical Process Parameter [CPP]**
**- Critical In-Process Control [CIPC]**

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# 3. Narrative on Unit Operations

### 3.1 Parameter Criticality Risk Assessment and PARs

As part of the process risk assessment, PARs are determined for any parameter designated as a 'potential risk' parameter. The majority of the PARs were mapped to ranges considered statistically improbable under common cause operational variability, or beyond, but not necessarily studied to the limits of failure for the process. Sigma (σ) has been defined as the routine operational variability for Lilly manufacturing equipment.

The Lilly 6σ ranges for processes were used. The resulting process operating parameter ranges were discussed with the contract manufacturer (CM) and confirmed that these would be practical ranges for equipment and operating control during the manufacturing of Eloralintide Fragment L2 Intermediate in the CM's equipment. If it was determined through historical data that the range for a parameter was already studied to an appropriate extent, and resulted in product meeting specification, then the result was documented and used as the PAR. For situations in which historical data were unavailable, experimental analyses were performed to establish PARs.

Based upon the results to the univariate and bracketing studies, the parameters identified in the risk assessments were classified as having a low, medium, or high risk of negatively impacting a CQA over the 6σ range. The resulting critical parameters and process controls are identified in Table 3.1.1 and 3.1.2.

**Table 3.1.1:** Eloralintide fragment L2 Critical Process Parameters (CPPs)

| Parameter | Type | PAR | Test Methodology | Potential Impact if PAR Exceeded |
|---|---|---|---|---|
| Resin loading - DIPEA quantity | CPP | 5.24 – **6.69 eq**. | N/A | Higher levels of DIPEA may lead to increased amounts of impurities. |
| Resin loading - Dissolution temperature | CPP | **17.0 – 33.8 °C** | N/A | Lower temperatures result in incomplete dissolution and activation of the AA, thus increased deletion impurities. Higher temperatures result in degradation impurities. |
| Resin loading – Coupling reaction temperature | CPP | **17.2 – 32.7 °C** | N/A | Lower temperatures result in reduced coupling thus increased deletion impurities. Higher temperatures result in degradation impurities. |
| Resin loading - Capping DMF wash quantity | CPP | **NLT 7.58 kg/kg** | N/A | Residual capping reagents can give rise to impurities in the subsequent step. |
| Deprotection temperature (Standard AA) | CPP | **17.0 – 34.8 °C** | N/A | Lower temperatures result in reduced deprotection thus increased deletion impurities. Higher temperatures result in degradation impurities. |
| Deprotection temperature (Pre-Ser20) | CPP | **14.6 – 34.7 °C** | N/A | Lower temperatures result in reduced deprotection thus increased deletion impurities. Higher temperatures result in degradation impurities. |
| Piperidine/DMF exposure time (Pre-Ser20 Treatment 1) | CPP | **6 – 68 min** | N/A | Short exposure times result in reduced deprotection thus increased deletion impurities. Longer exposure times result in degradation impurities. |
| Piperidine/DMF exposure time (Pre-Ser20 Treatment 2) | CPP | **6 – 83 min** | N/A | Short exposure times result in reduced deprotection thus increased deletion |

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| | | | | |
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| | | | | impurities. Longer exposure times result in degradation impurities. |
| Post deprotection IPAc wash quantity | CPP | **NLT 7.93 kg/kg** | N/A | Residual piperidine can increase in addition and acetyl-capped impurities |
| Oxyma quantity (Standard AA) | CPP | **1.94** – 2.25 **eq.** | N/A | Low Oxyma charge can cause DIC adduct and deletion impurities. |
| Oxyma quantity (NMeAsn22, αMePhe15) | CPP | **1.75** – 2.03 **eq.** | N/A | Low Oxyma charge can cause DIC adduct and deletion impurities. |
| Oxyma quantity (Asn21) | CPP | **2.50** – 2.93 **eq.** | N/A | Low Oxyma charge can cause DIC adduct and deletion impurities. |
| Dissolution jacket temperature (Standard AA) | CPP | **12.6 – 26.5 °C** | N/A | Lower temperatures result in incomplete dissolution and activation of the AA, thus increased deletion impurities. Higher temperatures result in degradation impurities. |
| Dissolution jacket temperature (Phe23) | CPP | **12.0 – 29.5 °C** | N/A | Lower temperatures result in incomplete dissolution and activation of the AA, thus increased deletion impurities. Higher temperatures result in degradation impurities. |
| Dissolution jacket temperature (NMeAsn22, αMePhe15) | CPP | **12.6 – 26.9 °C** | N/A | Lower temperatures result in incomplete dissolution and activation of the AA, thus increased deletion impurities. Higher temperatures result in degradation impurities. |
| Dissolution jacket temperature (Asn21) | CPP | **12.2 – 27.9 °C** | N/A | Lower temperatures result in incomplete dissolution and activation of the AA, thus increased deletion impurities. Higher temperatures result in degradation impurities. |
| DIC quantity (Standard AA) | CPP | **2.01** – 2.48 **eq.** | N/A | Low DIC charge can cause deletion impurities due to incomplete activation. |
| Activation temperature (Standard AA) | CPP | **12.6 – 26.5 °C** | N/A | Lower temperatures result in incomplete activation thus increased deletion impurities. Higher temperatures result in degradation impurities. |
| Activation temperature (Phe23) | CPP | **-2.5 – 13.6 °C** | N/A | Lower temperatures result in incomplete activation thus increased deletion impurities. Higher temperatures result in degradation impurities. |
| Activation temperature (NMeAsn22, αMePhe15) | CPP | **12.6 – 26.9 °C** | N/A | Lower temperatures result in incomplete activation thus increased deletion impurities. Higher temperatures result in degradation impurities. |
| Activation temperature (Asn21) | CPP | **12.2 – 27.9 °C** | N/A | Lower temperatures result in incomplete activation thus increased |

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| | | | | |
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| | | | | deletion impurities. Higher temperatures result in degradation impurities. |
| Coupling temperature (Standard AA) | CPP | **18.6 – 28.4 °C** | N/A | Lower temperatures result in reduced coupling thus increased deletion impurities. Higher temperatures result in degradation impurities. |
| Coupling temperature (Phe23) | CPP | **17.4 – 30.3 °C** | N/A | Lower temperatures result in reduced coupling thus increased deletion impurities. Higher temperatures result in degradation impurities. |
| Coupling temperature (NMeAsn22, αMePhe15) | CPP | **17.3 – 28.3 °C** | N/A | Lower temperatures result in reduced coupling thus increased deletion impurities. Higher temperatures result in degradation impurities. |
| Coupling temperature (Asn21) | CPP | **17.2 – 30.2 °C** | N/A | Lower temperatures result in reduced coupling thus increased deletion impurities. Higher temperatures result in degradation impurities. |
| Post Asn21 capping DMF wash quantity | CPP | **NLT 6.75 kg/kg** | N/A | Residual acetic anhydride and pyridine can cause acetyl capped impurities in the subsequent deprotection step. |
| IPAc wash quantity | CPP | **NLT 8.73 kg/kg** | N/A | Residual piperidine can cause removal of Fmoc group resulting in deFmoc impurity during drying/storage |
| Toluene wash quantity | CPP | **NLT 8.72 kg/kg** | N/A | Residual piperidine can cause removal of Fmoc group resulting in deFmoc impurity during drying/storage |
| MTBE wash quantity | CPP | **NLT 7.41 kg/kg** | N/A | Residual piperidine can cause removal of Fmoc group resulting in deFmoc impurity during drying/storage |

**Table 3.1.2:** Eloralintide fragment L2 In-Process Controls (IPCs)

| Parameter | Type | Acceptance criteria | Test Methodology | Potential Impact if PAR Exceeded |
|---|---|---|---|---|
| Water content prior to SPPS reaction | IPC | NMT 500 ppm | KF | Residual water interferes with the coupling reaction |
| Coupling completion colorimetric test | IPC | Pass (Negative x2) | Ninhydrin for Phe23, NMeAsn22, Ser20-αMePhe15, Ala13-Orn11; Chloranil for Asn21; Bromophenol Blue for Glu14 | Positive result indicates that coupling reaction has not occurred or is incomplete. |

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| | | | | |
|---|---|---|---|---|
| Post IPAc washing residual piperidine pre-Orn11 coupling | IPC | Pass (Negative) | Chloranil test comparing with 500 ppm piperidine in IPAc solution | Residual piperidine can cause removal of Fmoc group resulting in deFmoc and/or endo impurities in subsequent coupling step |
| Residual piperidine post final de-swelling MTBE wash | IPC | Pass (Negative) | Chloranil test comparing with 500 ppm piperidine in MTBE solution | Residual piperidine can cause removal of Fmoc group resulting in deFmoc impurity during drying/storage |
| SPPS Drying | IPC | NMT 9% w/w | LOD | Residual solvent specification. |

### 3.2 Process Overview

The process and its parameters discussed in this section are controlled through a combination of manual and automatic operations. Automation is coded in recipes that are used to perform each operation and may require manual input from operators through batch record instructions. Sampling and testing are also performed as noted in this section to confirm that steps perform as intended.

### *Solid Phase Peptide Synthesis*

The SPPS process can be broken down into five sub-steps: (1) CTC resin swelling; (2) Gly24 loading and capping; (3) Fmoc deprotection; (4) activation and coupling; and (5) de-swelling and drying. The critical process parameters and in-process controls for these sub-steps are listed in Table 3.1.1 and Table 3.1.2, respectively.

The 2-chlorotrityl chloride (2-CTC) resin is the limiting reagent and the basis for all SPPS charges. The resin charge quantity defines the SPPS batch scale (Refer to each CM Attachment). Solvent charges are expressed as weight ratios relative to the resin weight input (kg/kg). Amino acid and reagent charges are expressed as molar equivalents (eq) relative to the total moles of resin charged. Total resin moles are calculated as the actual resin weight charged (kg) multiplied by the actual resin substitution (mmol/g; specification 1.0 – 1.2 mmol/g) of each resin lot, summed where more than one lot is used. Fmoc-Gly-OH is charged in excess to load the resin to its maximum capacity. The subsequent amino acids and coupling reagents are also charged in molar excess to drive each coupling to completion.

**CTC Resin-Swelling**

Before the resin is charged, the activation tank and synthesizer are pre-treated with fresh DMF. The DMF is charged to the activation tank, stirred, transferred to the synthesizer and stirred under nitrogen. It is then sampled for water content [IPC] and drained. The pre-treatment is repeated until the water content meets the acceptance criterion (NMT 500 ppm). This limits the water introduced to the moisture-sensitive 2-CTC resin, because water hydrolyzes the active chlorotrityl chloride sites and reduces the achievable loading.

The target amount of 2-CTC resin is then charged to the synthesizer, followed by DMF, which is also used to rinse down the reactor wall. The resin is swelled in DMF with agitation and nitrogen bubbling for a fixed amount of time while controlling temperature. After swelling, the DMF is drained, and the resin is washed with DMF and drained in preparation for loading.

**Gly24 Loading & Capping**

The first amino acid, Fmoc-Gly-OH, is charged into the reactor with DMF and diisopropylethylamine (DIPEA). The reactor contents are stirred for a fixed time under nitrogen while maintaining a temperature to facilitate loading of Fmoc-Gly-OH onto the resin. DIPEA deprotonates Fmoc-Gly-OH and the resulting carboxylate reacts with CTC resin carbocation to form Fmoc-Gly-CTC resin and HCl. After loading, the reactor is drained, and the resulting Fmoc-Gly-CTC resin is washed and

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drained repeatedly with DMF. Each wash is first used to rinse the activation tank and transfer lines before being transferred to the synthesizer.

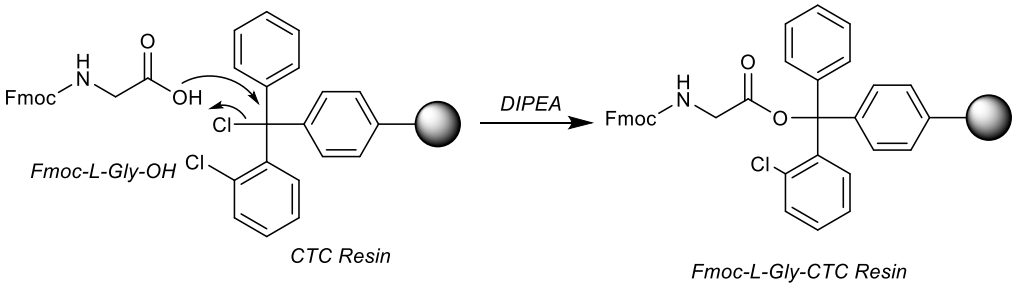



**Figure 3.2.1:** Loading Reaction Scheme

After washing, the resin is capped with the use of methanol in DMF solution and DIPEA to remove any unreacted active sites on the CTC resin from the loading step. For capping, the reactor's contents are stirred for a fixed time under nitrogen while controlling temperature. After capping, the reactor is drained and the Fmoc-Gly-CTC resin is washed multiple times with DMF.

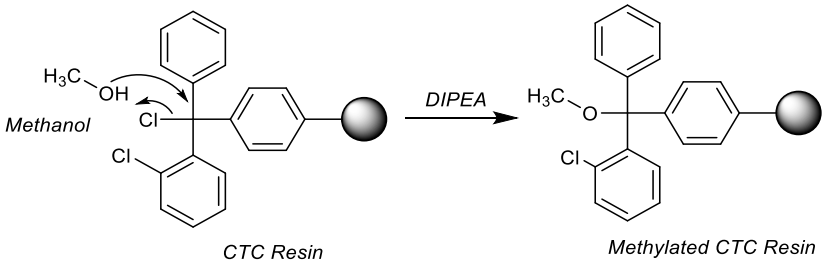



**Figure 3.2.2:** Capping Reaction Scheme

**Fmoc Deprotection**

The Fmoc-protected peptide on resin is deprotected using a 20% piperidine in DMF solution for a fixed time while stirring and controlling temperature, generating the free amine on the N-terminus. During this step, the liberated Fmoc decomposes into $CO_2$ and dibenzofulvene (DBF), which is scavenged by piperidine to form a stable adduct. Free DBF can otherwise undergo free-radical polymerization, forming a gel-like material if not adequately scavenged. After deprotection, the reactor is drained and the peptide-resin is washed multiple times with IPAc to remove the DBF–piperidine adduct and excess piperidine followed by DMF washes to remove residual IPAc and to re-swell the resin. The DMF washes also remove any acetic acid carried in from the IPAc, which would otherwise acetylate the free N-terminus and form truncated (acetyl-capped) sequences. Prior to the Orn11 coupling, and extended IPAc wash is applied, and a chloranil test is carried out to check if residual piperidine has been washed away to NMT 500 ppm before continuing to the DMF re-swell step. In the case of a failing IPC, more IPAc washes are carried out, and the IPC test is repeated.

Coupling steps should follow immediately after each preceding deprotection to minimize side reactions of the free amine. Deprotection is intentionally not performed directly after Asn21 (Fmoc-L-Asn(Trt)-OH). Instead, an acetic anhydride/pyridine capping step is used to mitigate DKP formation and to prevent Ac-end-capping 4-mer and DIC-end-capping 3-mer impurities. Additionally, deprotection is not required after the Orn11 coupling, as it is the final amino acid in the sequence

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**Chloranil wash completion test**

The chloranil test is used as an IPC to detect unprotected secondary amines and, in some cases, primary amines present on the resin. A positive chloranil test is indicated by the formation of a blue-green color, indicating that there is still piperidine present from the deprotection and that the washing is incomplete, while a negative result indicates a successful wash.

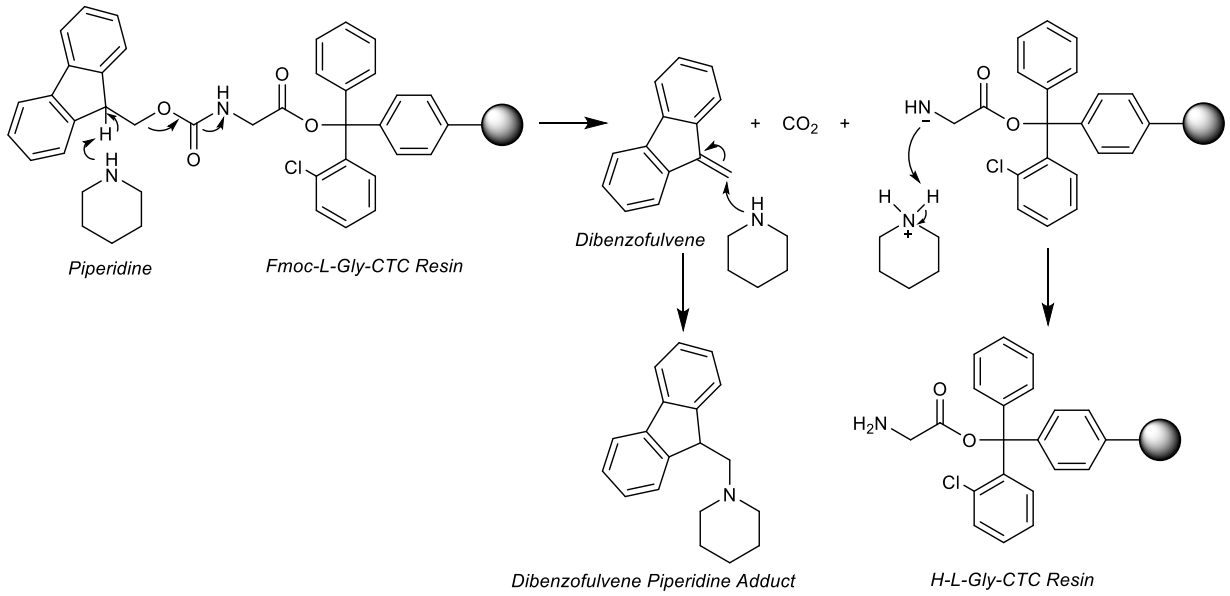



**Figure 3.2.3:** Fmoc Deprotection Reaction Scheme Example

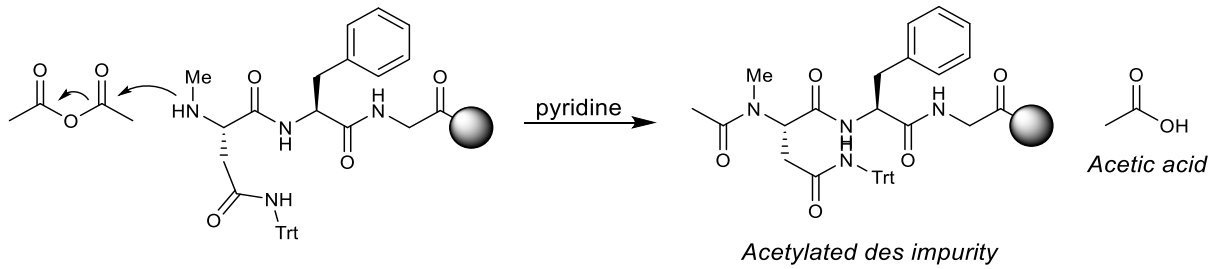



**Figure 3.2.4:** Capping reaction scheme

**Activation and Coupling Phe23**

For coupling of Phe23, a target amount of the starting material is dissolved in DMF in a separate vessel and cooled while stirring. After dissolution, the Fmoc-L-Phe-OH is activated under nitrogen for a fixed amount of time using PyBOP, DIPEA and Oxyma Pure while controlling temperature. The chilled solution is then added to the reactor containing the deprotected resin and allowed to react for a fixed amount of time while controlling temperature.

At designated coupling times, samples are obtained and tested using the qualitative ninhydrin (i.e. Kaiser) IPC method to determine reaction completion. If the reaction is incomplete, as indicated by the presence of a blue color, the reaction is extended. For the IPC to pass, two consecutive samples have to be negative. Following the coupling the reactor is drained, and the peptide-resin is washed with DMF.

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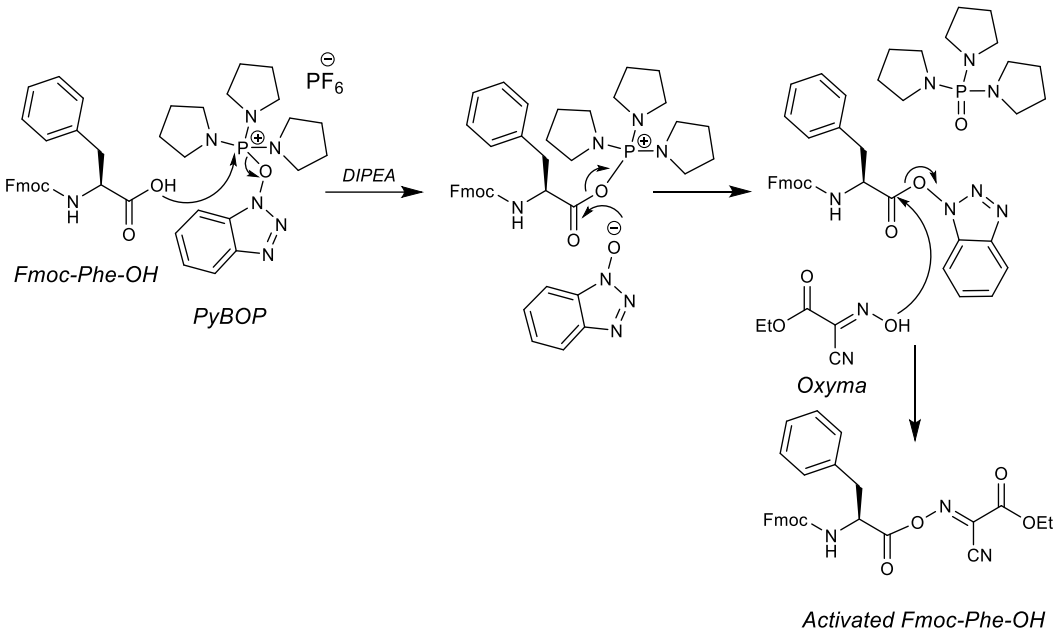



**Figure 3.2.5:** Phe23 coupling using PyBOP

**Activation & Coupling (N-Me-Asn22/Asn21/Ser20/Ser19/Arg18/Val17/Leu16/αMePhe15/Glu14/Ala13/Leu12/Orn11)**

For the remaining amino acid coupling cycles, activation is first performed using target amounts of amino acid derivative, Oxyma, DIC, and DMF. Each amino acid is activated for a fixed amount of time while controlling temperature. The acid moiety of the amino acid reacts with 1,3-diisopropylcarbodiimide (DIC) to form an activated ester. This ester further reacts with Oxyma to generate the Oxyma activated ester, a more stable ester that is resistant to racemization. Diisopropylurea (DIU) is generated as a byproduct and may precipitate in the process. Though it has enough solubility to theoretically dissolve during the post-coupling washes, there is a risk that the crystallized DIU will negatively impact filtration. A multitude of unintended side-reactions may occur during activation.[1]

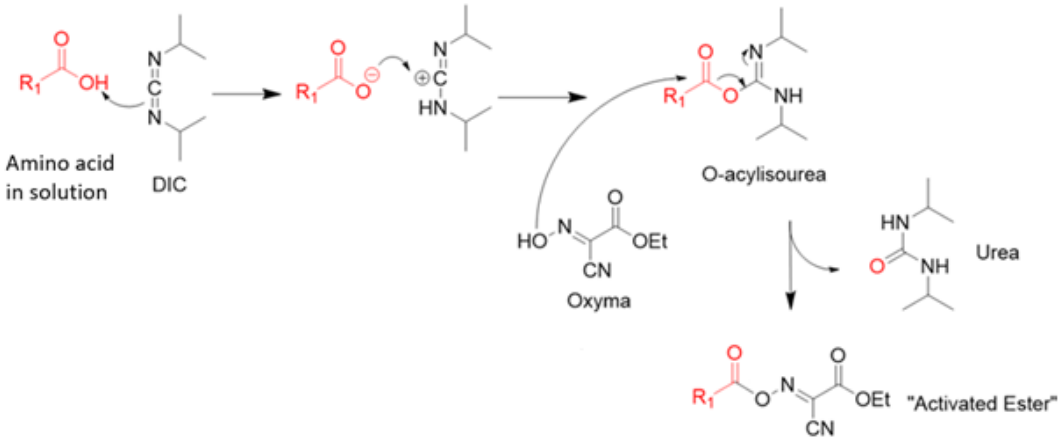



**Figure 3.2.6:** Amino Acid Activation Reaction Scheme

[1] Development of a Mechanistic Amino Acid Activation Reaction Kinetics Model for Safe and Efficient Solid Phase Peptide Synthesis; J. Wang et al., *Organic Process Research & Development 2023 62 (37), 14986-14996*.

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The resulting activated solution is transferred to the SPPS reactor under nitrogen to commence coupling. Each coupling is performed for a fixed amount of time while controlling temperature. The activated amino acid esters readily couple to the free amine of the growing peptide chain. Oxyma is regenerated in the process of coupling.

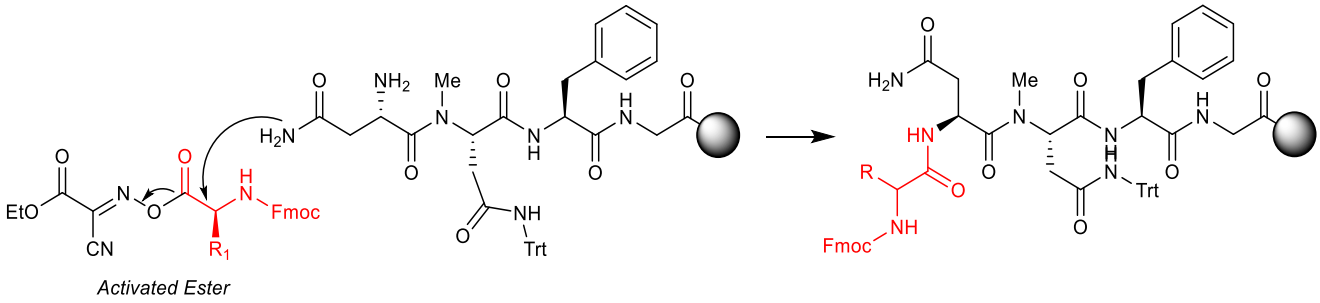



**Figure 3.2.7:** Amino Acid Coupling Reaction Scheme

After designated coupling times, samples are obtained and tested using the qualitative ninhydrin (i.e., Kaiser) (NMeAsn22, Ser20-αMePhe15 and Ala13-Orn11), chloranil (Asn21) or Bromophenol Blue (Glu14) IPC method to determine reaction completion. If the reaction is incomplete, as indicated by the presence of a blue (ninhydrin, Bromophenol Blue) or blue-green (Asn21) color, the reaction is extended. For Asn21, kicker charges of Oxyma and DIC can be added after a certain period of time to push the reaction to completion. For the IPC to pass, two consecutive samples have to be negative. Following the coupling, the reactor is drained and the peptide-resin is washed with DMF, with the exception of completion of the coupling of Orn11, after which the peptide-resin deswell is started, using nitrogen bubbling and controlled agitation, to remove piperidine, activated amino-acid solutions, and soluble byproducts..

The Fmoc protecting group is removed prior to each coupling step, utilizing the deprotection procedure described previously.

**Kaiser, Chloranil and Bromophenol Blue coupling completion testing**

The Kaiser (ninhydrin) test is used as an IPC following amino acid coupling steps to assess coupling completeness. The test detects the presence of free primary amines on the resin, which indicate incomplete coupling. When free amines are present, reaction with ninhydrin produces a blue or purple chromophore, corresponding to a positive (failed) test result. A negative test result, indicating successful coupling, is characterized by no color change or a light yellow to brown coloration.

The chloranil test is used as an IPC to detect unprotected secondary amines and, in some cases, primary amines present on the resin. A positive chloranil test is indicated by the formation of a blue-green color, indicating that the coupling is incomplete, while a negative result indicates a successful coupling.

A Bromophenol Blue test can also be used as an IPC to determine whether the coupling has been successful. The coupling is complete when two consecutive samples test negative (yellow to orange resin). A blue to black color indicates unreacted free amine and an incomplete coupling reaction.

**Peptide-Resin**

The peptidyl resin is dewelled in a defined sequence by first washing multiple times with IPAc, using nitrogen bubbling and controlled agitation, to remove DMF. The resin is then washed with toluene followed by MTBE, and then another cycle of toluene and MTBE washes, completing deswelling and removing remaining higher-boiling solvents. Each wash is performed with controlled mixing and full drainage. As sample is taken from the final MTBE wash and analysed using the chloranil method. If the sample fails, the toluene and MTBE wash cycle is repeated and another sample is analysed until the chloranil endpoint has been met..

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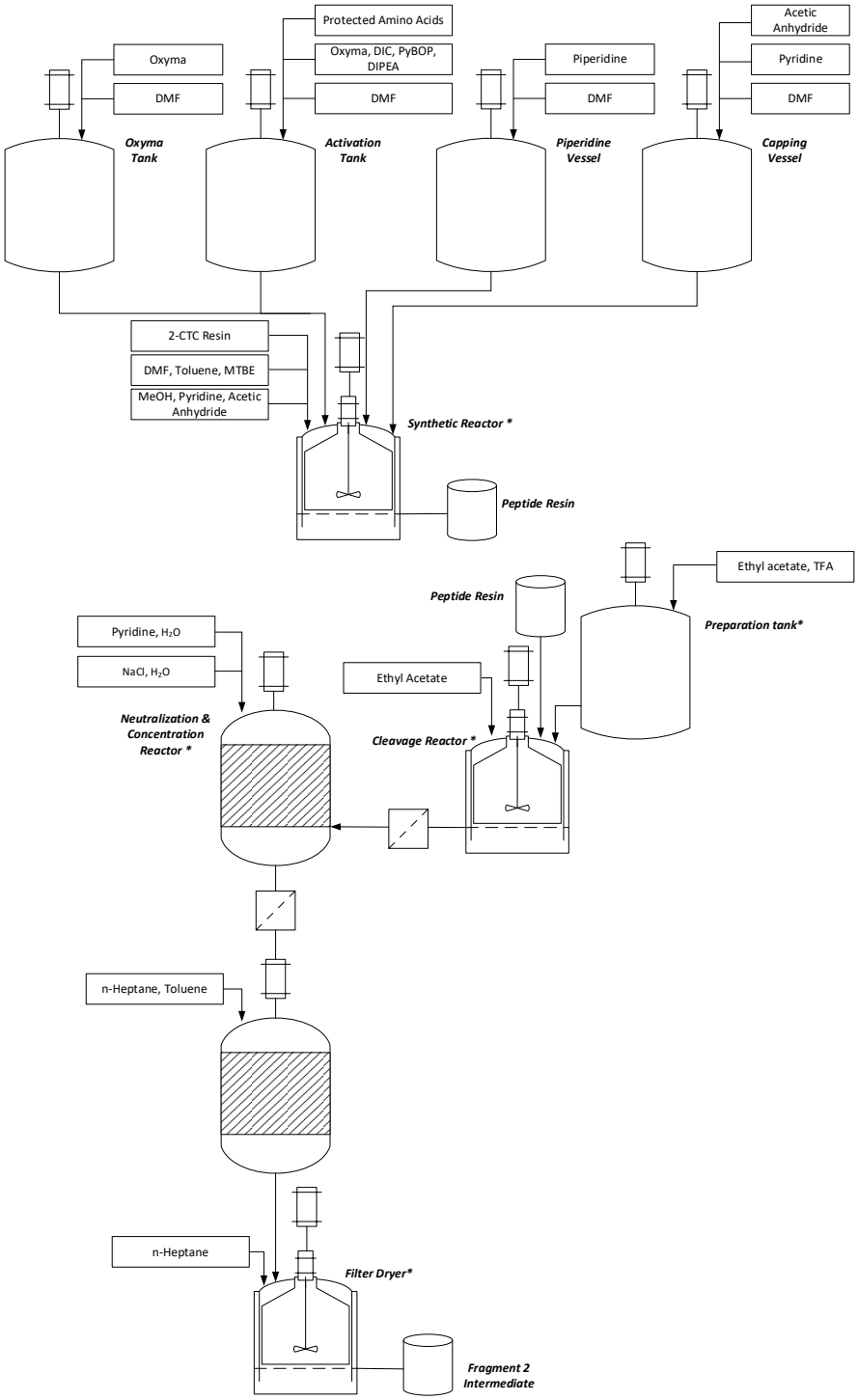
Following solvent washes, the resin is dried under nitrogen sweep and vacuum at controlled temperatures until the LOD specification is met. The dried material is collected into double LDPE bags, weighed, labeled, and placed into drums for temporary storage.

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## 4. Equipment Flow Chart (EFC)

This section demonstrates a simplified schematic that shows the major pieces of equipment. Individual CM's may have slightly different configurations, which are outlined in more detail in the site-specific attachment documents.




*Vessels noted with specific materials of construction (e.g., glass-lined steel, Hastelloy®, or other corrosion-resistant alloys) are required to ensure compatibility with strong acids, halogenated organics, and other chemically aggressive reagents used throughout the process. These materials provide resistance to corrosion, minimize extractables/leachables, and maintain GMP-compliant product purity.

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# 5. Material Specification Rationale

Detailed specification rationale and its link to CQAs are documented in the specification rationale document. Impurity spiking studies were conducted with starting material impurities to verify no impact to LAARA Fragment L2 Intermediate. In addition, historical data from developmental batches manufactured prior to validation were factored into setting the proposed specifications for starting materials and LAARA Fragment L2 Intermediate. The specification rationale for raw material and solvent specifications was based on the demonstrated performance of the materials controlled by the contract manufacturers' specifications during development.

## 5.1 Raw Material Specifications

The raw materials and solvents used in the synthesis of Eloralintide Fragment L2 are outlined in Table 5.1.

**Table 5.1:** Raw Material Specifications

| Raw Material | Method | Acceptance Criteria |
|---|---|---|
| 2-Chlorotrityl Chloride Resin * | Substitution degree | 1.0 – 1.2 mmol/g |
| | Swelling in DMF | 3.0 – 6.0 ml/g |
| | Particle size (D90) | 105 – 180 um |
| Acetic anhydride | Purity | NLT 99.0% |
| Benzotriazole-1-yl-oxy-tris-pyrrolidino-phosphonium Hexafluorophosphate (PyBOP) | Purity | NLT 99.0% |
| 1,3-Diisopropylcarbodiimide (DIC) | Purity | NLT 99.0% |
| *N*,*N*-Diisopropylethylamine | Purity | NLT 99.0% |
| *N*,*N*-Dimethylformamide | Purity | NLT 99.0% |
| Ethyl acetate | Purity | NLT 99.0% |
| Ethyl cyanohydroxyiminoacetate (Oxyma) | Purity | NLT 99.0% |
| *n*-Heptane | Purity | NLT 99.0% |
| Isopropyl acetate | Purity | NLT 99.0% |
| Methanol | Purity | NLT 99.0% |
| Methyl tert-Butyl Ether | Purity | NLT 99.0% |
| Piperidine | Purity | NLT 97.0% |
| Pyridine | Purity | NLT 99.0% |
| Sodium chloride | Assay | NLT 99.0% |
| Toluene | Purity | NLT 99.0% |
| Trifluoroacetic acid | Assay | NLT 98.0% |
| Purified Water | USP, Ph.Eur | Pass** |

* Critical raw material.

** Purified water quality is compliant with current USP, Ph.Eur.

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## 5.2 Starting Material Specifications

The specifications for the registered starting materials that are utilized in the manufacture of Eloralintide Fragment L2 are summarized below, see specification rationale document for regulatory starting materials.

**Table 5.2.1:** Starting Material Specifications

| Starting Material | Method | Acceptance Criteria |
|---|---|---|
| Fmoc-α-Me-Phe-OH | Visual Appearance | White to off-white powder |
| | Identification (by $^1$H NMR or HPLC) | $^1$H NMR: Conforms to structure<br>HPLC: Retention time confirms to reference |
| | Purity by HPLC (% area) | ≥99.0 |
| | Fmoc-D-α-Me-Phe-OH by HPLC (% area) | ≤0.2 |
| | Individual Unspecified Impurity by HPLC (% area) | ≤0.2 |
| | H-α-Me-Phe-OH by HPLC (% w/w) | ≤0.1 |
| | Water content (% w/w) | ≤1.0 |
| | Assay by HPLC (% w/w, as-is) | ≥97.0 |
| Fmoc-L-Ala-OH ·$H_2O$ | Visual Appearance | White to off-white powder |
| | Identification (by $^1$H NMR or HPLC) | $^1$H NMR: Conforms to structure<br>HPLC: Retention time confirms to reference |
| | Purity by HPLC (% area) | ≥99.0 |
| | Fmoc-β-Ala-OH by HPLC (% area) | ≤0.2 |
| | Fmoc-β-Ala-Ala-OH by HPLC (% area) | ≤0.2 |
| | Fmoc-Ala-Ala-OH by HPLC (% area) | ≤0.2 |
| | Fmoc-D-Ala-OH by HPLC (% area) | ≤0.2 |
| | Individual Unspecified Impurity by HPLC (% area) | ≤0.2 |
| | H-Ala-OH by HPLC (% w/w) | ≤0.1 |
| | Water content (% w/w) | ≤6.0 |
| | Assay by HPLC (% w/w, as-is) | ≥97.0 |
| Fmoc-L-Arg(Pbf)-OH | Visual Appearance | White to off-white powder |
| | Identification (by $^1$H NMR or HPLC) | $^1$H NMR: Conforms to structure<br>HPLC: Retention time confirms to reference |
| | Purity by HPLC (% area) | ≥99.0 |
| | Fmoc-β-Ala-OH by HPLC (% area) | ≤0.2 |
| | Fmoc-D-Arg(Pbf)-OH by HPLC (% area) | ≤0.2 |
| | Individual Unspecified Impurity by HPLC (% area) | ≤0.2 |
| | H-Arg(Pbf)-OH by HPLC (% w/w) | ≤0.1 |
| | Water content (% w/w) | ≤1.0 |
| | Assay by HPLC (% w/w, as-is) | ≥97.0 |

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| | | |
|---|---|---|
| Fmoc-L-Asn(Trt)-OH | Visual Appearance | White to off-white powder |
| | Identification (by $^1$H NMR or HPLC) | $^1$H NMR: Conforms to structure<br>HPLC: Retention time confirms to reference |
| | Purity by HPLC (% area) | ≥99.0 |
| | Fmoc-β-Ala-OH by HPLC (% area) | ≤0.2 |
| | H-Asn(Trt)-OH by HPLC (% area) | ≤0.1 |
| | Fmoc-D-Asn(Trt)-OH by HPLC (% area) | ≤0.2 |
| | Individual Unspecified Impurity by HPLC (% area) | ≤0.2 |
| | Water content (% w/w) | ≤1.0 |
| | Assay by HPLC (% w/w, as-is) | ≥97.0 |
| Fmoc-L-Glu(OtBu)-OH·$H_2O$ | Visual Appearance | White to off-white powder |
| | Identification (by $^1$H NMR or HPLC) | $^1$H NMR: Conforms to structure<br>HPLC: Retention time confirms to reference |
| | Purity by HPLC (% area) | ≥99.0 |
| | Fmoc-β-Ala-OH by HPLC (% area) | ≤0.2 |
| | Fmoc- β-Ala-Glu(OtBu)-OH by HPLC (% area) | ≤0.2 |
| | Fmoc-D-Glu(OtBu)-OH by HPLC (% area) | ≤0.2 |
| | Individual Unspecified Impurity by HPLC (% area) | ≤0.2 |
| | H-Glu(OtBu)-OH (% w/w) | ≤0.1 |
| | Water content (% w/w) | ≤6.0 |
| | Assay by HPLC (% w/w, as-is) | ≥97.0 |
| Fmoc-Gly-OH | Visual Appearance | White to off-white powder |
| | Identification (by $^1$H NMR or HPLC) | $^1$H NMR: Conforms to structure<br>HPLC: Retention time confirms to reference |
| | Purity by HPLC (% area) | ≥99.0 |
| | Fmoc-β-Ala-Gly-OH by HPLC (% area) | ≤0.2 |
| | Fmoc-Gly-Gly-OH by HPLC (% area) | ≤0.1 |
| | Individual Unspecified Impurity by HPLC (% area) | ≤0.2 |
| | H-Gly-OH (% w/w) | ≤0.1 |
| | Water content (% w/w) | ≤1.0 |
| | Assay by HPLC (% w/w, as-is) | ≥97.0 |
| Fmoc-L-Leu-OH | Visual Appearance | White to off-white powder |
| | Identification (by $^1$H NMR or HPLC) | $^1$H NMR: Conforms to structure<br>HPLC: Retention time confirms to reference |
| | Purity by HPLC (% area) | ≥99.0 |
| | Fmoc-β-Ala-OH by HPLC (% area) | ≤0.2 |

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| | | |
|---|---|---|
| | Fmoc-L-Ile-OH by HPLC (% area) | ≤0.1 |
| | Fmoc-D-Leu-OH by HPLC (% area) | ≤0.2 |
| | Individual Unspecified Impurity by HPLC (% area) | ≤0.2 |
| | H-Leu-OH (% w/w) | ≤0.1 |
| | Water content (% w/w) | ≤1.0 |
| | Assay by HPLC (% w/w, as-is) | ≥97.0 |
| Fmoc-L-N-Me-Asn(Trt)-OH | Visual Appearance | White to off-white powder |
| | Identification (by $^1$H NMR or HPLC) | $^1$H NMR: Conforms to structure<br>HPLC: Retention time confirms to reference |
| | Purity by HPLC (% area) | ≥99.0 |
| | Fmoc-β-Ala-OH by HPLC (% area) | ≤0.2 |
| | Fmoc-L-Asn(Trt)-OH by HPLC (% area) | ≤0.1 |
| | Fmoc-D-N-Me-Asn(Trt)-OH (% area) | ≤0.2 |
| | Individual Unspecified Impurity by HPLC (% area) | ≤0.2 |
| | H-N-Me-Asn(Trt)-OH (% w/w) | ≤0.1 |
| | Water content (% w/w) | ≤2.0 |
| | Assay by HPLC (% w/w, as-is) | ≥97.0 |
| Fmoc-L-Orn(Boc)-OH | Visual Appearance | White to off-white powder |
| | Identification (by $^1$H NMR or HPLC) | $^1$H NMR: Conforms to structure<br>HPLC: Retention time confirms to reference |
| | Purity by HPLC (% area) | ≥99.0 |
| | Fmoc-β-Ala-OH by HPLC (% area) | ≤0.2 |
| | Fmoc-D-Orn(Boc)-OH by HPLC (% area) | ≤0.2 |
| | Individual Unspecified Impurity by HPLC (% area) | ≤0.2 |
| | H-Orn(Boc)-OH (% w/w) | ≤0.1 |
| | Water content (% w/w) | ≤1.0 |
| | Assay by HPLC (% w/w, as-is) | ≥97.0 |
| Fmoc-L-Phe-OH | Visual Appearance | White to off-white powder |
| | Identification (by $^1$H NMR or HPLC) | $^1$H NMR: Conforms to structure<br>HPLC: Retention time confirms to reference |
| | Purity by HPLC (% area) | ≥99.0 |
| | Fmoc-β-Ala-OH by HPLC (% area) | ≤0.2 |
| | Fmoc-D-Phe-OH by HPLC (% area) | ≤0.2 |
| | Individual Unspecified Impurity by HPLC (% area) | ≤0.2 |
| | H-Phe-OH (% w/w) | ≤0.1 |
| | Water content (% w/w) | ≤1.0 |
| | Assay by HPLC (% w/w, as-is) | ≥97.0 |

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| | | |
|---|---|---|
| Fmoc-L-Ser(tBu)-OH | Visual Appearance | White to off-white powder |
| | Identification (by $^1$H NMR or HPLC) | $^1$H NMR: Conforms to structure<br>HPLC: Retention time confirms to reference |
| | Purity by HPLC (% area) | ≥99.0 |
| | Fmoc-β-Ala-OH by HPLC (% area) | ≤0.2 |
| | Fmoc-D-Ser(tBu)-OH by HPLC (% area) | ≤0.2 |
| | Individual Unspecified Impurity by HPLC (% area) | ≤0.2 |
| | H-Ser(tBu)-OH (% w/w) | ≤0.1 |
| | Water content (% w/w) | ≤1.0 |
| | Assay by HPLC (% w/w, as-is) | ≥97.0 |
| Fmoc-L-Val-OH | Visual Appearance | White to off-white powder |
| | Identification (by $^1$H NMR or HPLC) | $^1$H NMR: Conforms to structure<br>HPLC: Retention time confirms to reference |
| | Purity by HPLC (% area) | ≥99.0 |
| | Fmoc-β-Ala-OH by HPLC (% area) | ≤0.2 |
| | Fmoc-Leu-OH by HPLC (% area) | ≤0.1 |
| | Fmoc-Ile-OH by HPLC (% area) | ≤0.1 |
| | Fmoc-D-Val-OH by HPLC (% area) | ≤0.2 |
| | Individual Unspecified Impurity by HPLC (% area) | ≤0.2 |
| | H-Val-OH (% w/w) | ≤0.1 |
| | Water content (% w/w) | ≤1.0 |
| | Assay by HPLC (% w/w, as-is) | ≥97.0 |

The following starting material manufacturers have been approved for use in LAARA Fragment 2.

**Table 5.2.2:** Starting Material Manufacturers

| Starting Material | Manufacturer | Site Address |
|---|---|---|
| **Protected Amino Acid Derivatives** | | |
| Fmoc-Gly-OH | Asymchem DH3 | Jilin Asymchem Laboratories Co., Ltd.<br>No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin, 133700, China |
| | Asymchem FX2 | Liaoning Asymchem Laboratories Co., Ltd.<br>Yimatu village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129, China |
| | ZY Biochem | Chengdu ZY Biochemical Technology Co., Ltd<br>No. 170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, China |

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| | | |
|---|---|---|
| Fmoc-L-Ala-OH·$H_2O$ | Asymchem DH3 | Jilin Asymchem Laboratories Co., Ltd.<br>No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin, 133700, China |
| | Asymchem FX2 | Liaoning Asymchem Laboratories Co., Ltd.<br>Yimatu village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129, China |
| | ZY Biochem | Chengdu ZY Biochemical Technology Co., Ltd<br>No. 170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, China |
| Fmoc-L-Arg(Pbf)-OH | Asymchem DH3 | Jilin Asymchem Laboratories Co., Ltd.<br>No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin, 133700, China |
| | ZY Biochem | Chengdu ZY Biochemical Technology Co., Ltd<br>No. 170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, China |
| Fmoc-L-Asn(Trt)-OH | Asymchem DH3 | Jilin Asymchem Laboratories Co., Ltd.<br>No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin, 133700, China |
| | Asymchem FX2 | Liaoning Asymchem Laboratories Co., Ltd.<br>Yimatu village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129, China |
| | ZY Biochem | Chengdu ZY Biochemical Technology Co., Ltd<br>No. 170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, China |
| Fmoc-L-Glu(OtBu)-OH·$H_2O$ | Asymchem DH3 | Jilin Asymchem Laboratories Co., Ltd.<br>No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin, 133700, China |
| | ZY Biochem | Chengdu ZY Biochemical Technology Co., Ltd<br>No. 170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, China |
| Fmoc-L-Leu-OH | Asymchem DH3 | Jilin Asymchem Laboratories Co., Ltd.<br>No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin, 133700, China |
| | Asymchem FX2 | Liaoning Asymchem Laboratories Co., Ltd.<br>Yimatu village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129, China |

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| | | |
|---|---|---|
| | ZY Biochem | Chengdu ZY Biochemical Technology Co., Ltd<br>No. 170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, China |
| Fmoc-L-Phe-OH | Asymchem DH3 | Jilin Asymchem Laboratories Co., Ltd.<br>No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin, 133700, China |
| | Asymchem FX2 | Liaoning Asymchem Laboratories Co., Ltd.<br>Yimatu village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129, China |
| | ZY Biochem | Chengdu ZY Biochemical Technology Co., Ltd<br>No. 170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, China |
| Fmoc-L-Ser(tBu)-OH | Asymchem DH3 | Jilin Asymchem Laboratories Co., Ltd.<br>No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin, 133700, China |
| | Asymchem FX2 | Liaoning Asymchem Laboratories Co., Ltd.<br>Yimatu village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129, China |
| | ZY Biochem | Chengdu ZY Biochemical Technology Co., Ltd<br>No. 170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, China |
| Fmoc-L-Val-OH | Asymchem DH3 | Jilin Asymchem Laboratories Co., Ltd.<br>No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin, 133700, China |
| | Asymchem FX2 | Liaoning Asymchem Laboratories Co., Ltd.<br>Yimatu village, Yimatu town, Fu Mongolia county (Fluorine Industry Development Area), Fuxin, Liaoning, 123129, China |
| | ZY Biochem | Chengdu ZY Biochemical Technology Co., Ltd<br>No. 170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, China |
| **Non-coded Amino Acids** | | |
| Fmoc-L-Orn(Boc)-OH | Asymchem DH3 | Jilin Asymchem Laboratories Co., Ltd.<br>No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin, 133700, China |

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| | | |
|---|---|---|
| | ZY Biochem | Chengdu ZY Biochemical Technology Co., Ltd<br>No. 170 Qingpu Road, Shou'an Subdistrict, Pujiang County, Chengdu City, Sichuan Province, China |
| Fmoc-N-MeAsn(Trt)-OH | Asymchem DH3 | Jilin Asymchem Laboratories Co., Ltd.<br>No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin, 133700, China |
| Fmoc-α-Me-Phe-OH | Asymchem DH3 | Jilin Asymchem Laboratories Co., Ltd.<br>No. 899, Xingqi Road, High-tech Industrial Zone, Dunhua, Jilin, 133700, China |
| **Resin** | | |
| 2-CTC-Resin* | Hecheng | Tianjin Nankai Hecheng S&T Co., Ltd.<br>969-6, Kaixuan Street, Dagang Petrochemical Industrial Park, Binhai New District, Tianjin, China |
| | Sunresin | Sunresin New Materials Co, Ltd. Gao ling<br>No. 135, Jinye Road, Xi'an Hitech Industrial Development Zone, Shaanxi-710076, China |

* Critical raw material

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## 5.3 Product Specifications

The specifications for Eloralintide Fragment L2 intermediate, which can be found in the specification rationale document, are summarized below.

**Table 5.3:** Eloralintide Fragment L2 Product Specifications

| Test | Analytical Procedure | Acceptance Criteria |
|---|---|---|
| **Identification Test** | | |
| Identity | UHPLC-UV | Retention time compares favorably with that of the reference standard. |
| **Potency Test** | | |
| Assay | UHPLC-UV | Not less than 80.0% w/w |
| **Purity Tests** | | |
| LSN4182924 Main Peak Purity Impurities | UHPLC-UV | Not less than 90.0% area |
| Total | UHPLC-UV | Not more than 10.0% area |
| L2 des-(Asn21-NMeAsn22) (LSN5973627) | UHPLC-UV (RRT 0.78) | Not more than 2.0% area |
| L2 endo-Gly24 (LSN5973649) | UHPLC-UV (RRT 0.96) | Not more than 0.5% area |
| Any Unspecified Impurity | UHPLC-UV | Not more than 0.5% area |
| **Other Tests** | | |
| Description | Visual | White to yellow solid |

## 5.4 Product Packaging and Storage

The final Eloralintide Fragment L2 intermediate material is packaged in a polyethylene bag, which has a foil overwrap that is placed inside a high-density polyethylene drum. The packaged product is stored and shipped at 2 to 8 °C. Packaging and storage details for the Fragment L2 product are shown in Table 5.5.

**Table 5.4:** Packaging and Storage Details

| Step | Container/Packaging | Specifications | Storage | Shipping |
|---|---|---|---|---|
| Peptide-Resin | Primary | • Polyethylene terephthalate drying bag or polyethylene bag | 2 – 8 °C | N/A |
| | Secondary | • Antistatic low-density polyethylene (LDPE) bag in fibre drum, if primary is drying bag<br>• High-density polyethylene drum, if primary is polyethylene bag | | |

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# 6. Stability & Lifetimes

## 6.1 Stable Hold Points

The following "stable hold points" have been identified in the manufacture of LAARA Fragment L2 intermediate. In each case it has been demonstrated that there is no impact on quality by holding the process at certain points for the extended period of time given.

**Table 6.1:** LAARA Fragment L2 Stable Hold Points

| Step | Operation | Conditions (°C)* | Hold Time |
|---|---|---|---|
| Step 1 SPPS | Coupling long hold – Gly24 | 24.7 | NMT 480 min |
| | Coupling long hold – Phe23 | 22.3 | NMT 841 min |
| | Coupling long hold – NMeAsn22 | 24.1 | NMT 841 min |
| | Coupling long hold – Asn21 | 22.1 | NMT 1680 min |
| | Coupling long hold – Ser20 | 22.4 | NMT 600 min |
| | Coupling long hold – Ser19 | 22.1 | NMT 600 min |
| | Coupling long hold – Arg18 | 22.3 | NMT 600 min |
| | Coupling long hold – Val17 | 22.3 | NMT 600 min |
| | Coupling long hold – Leu16 | 22.4 | NMT 600 min |
| | Coupling long hold – αMePhe15 | 22.2 | NMT 840 min |
| | Coupling long hold – Glu14 | 22.3 | NMT 1080 min |
| | Coupling long hold – Ala13 | 22.2 | NMT 600 min |
| | Coupling long hold – Leu12 | 22.1 | NMT 600 min |
| | Coupling long hold – Orn11 | 22.3 | NMT 600 min |
| | Fmoc-Fragment 2 on resin – drying | 42 | 48 h |
| | Fmoc-Fragment 2 on resin – long-term storage | 2-8 | 7 months |

* In the PAR experiments, the coupling start time was recorded when the coupling temperature reached within 3 °C of the set point. Therefore, although the reported temperature appears lower than the baseline value, it reflects a limitation of the recorded value; the temperature most likely reached the set point within the hold time.

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## 7. Impurity Identification & Control

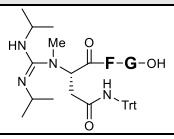
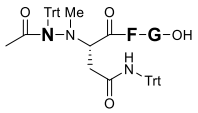
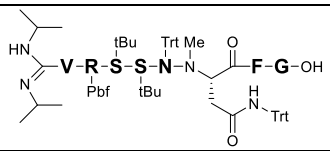
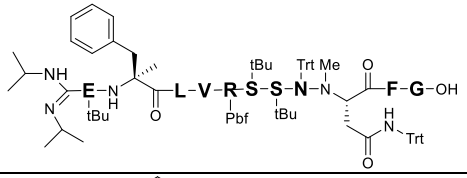
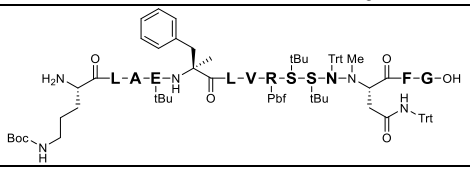
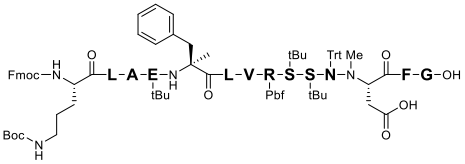
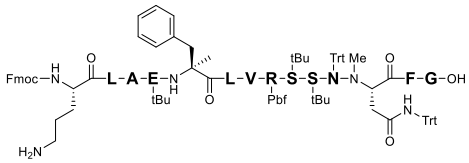
The process control strategy was designed to address the following impurities. A process risk assessment was performed to address these potential impurities and experimental testing was conducted to ascertain their formation and control points, see specification rationale document.

**Table 7.1:** LAARA Fragment L2 Impurity Types & Controls

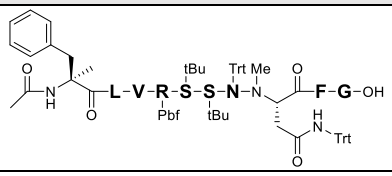
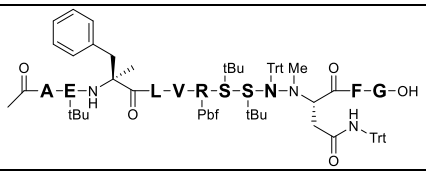
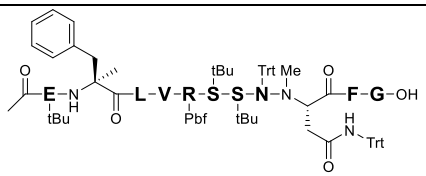
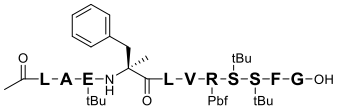
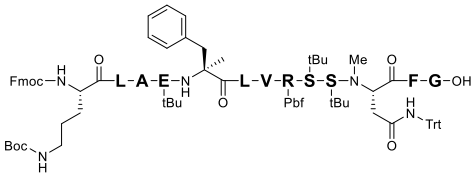
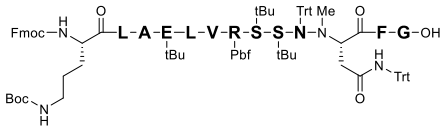
| CQA | Impurity Source/Formation | Control Strategy |
|---|---|---|
| Addition (endo) Impurities | Can form during synthesis if Fmoc protecting group is removed from the coupling AA, allowing a second identical residue to couple. This can occur if piperidine is not suitably washed out of the resin. Addition impurities can also form if the amino acid starting material has an appreciable amount of non-Fmoc protected material or if the Fmoc amino acid contains Fmoc di-peptide. | Addition impurities are controlled through starting material specifications and process parameters in post deprotection washes to ensure adequate piperidine removal. 2-CTC resin with known addition impurity risk is capped after 1$^{st}$ coupling. |
| Deletion (des) Impurities | These are the most likely group of impurities to form during the SPPS build process. These impurities can form during synthesis if the coupling reaction is not complete before moving to the next step or if the preceding piperidine-mediated deprotection did not go to completion. | Implementing an IPC for end of the coupling reaction. Substantial number of deletion impurities that are not cleared through chromatography purification are early build cycle deletions. |
| Acetyl-capped Impurities | These impurities can form when acetic acid impurity in IPAc or in AAs are carried into the coupling steps. | Parametric/procedural controls by DMF reswelling wash following IPAc wash. Acetic acid is controlled in starting material supplier specifications. |
| Racemization (i.e. *D*-isomers) | The *D*-isomer form of the amin acid can be formed in the build synthesis during the solution phase activation, followed by coupling into the peptide. *D*-isomers may also be present in the starting amino acid material. | *D*-isomers are primarily controlled through raw material specifications and process parameters. Cold activation and coupling for racemization prone coupling cycles. |
| Substitution Impurities | Substitution impurities may form if a different amino acid is present in the reaction mixture during the coupling phase, which may occur either from incomplete washing or an impurity in the starting raw material. Substitution impurities have not been observed to a significant level, outside of those related to raw material impurities (often related to β-alanine impurity). | Substitution impurities may be controlled by ensuring sufficient washing is conducted after each cycle; however, the primary point of substitution impurity control is raw material specifications. |
| Protecting group modification/loss | These impurities can form if the protecting group on a secondary functional group is removed during synthesis and an alternate reaction occurs. In Fragment L2, examples of these impurities include the loss of the Boc protecting group for Orn11, the loss of the $^{t}$Bu protecting group for Glu14, or the loss of the Trt protecting group for Asn21 or NMeAsn22. | Established PARs and well-controlled processing conditions ensure that these impurity levels are minimized. |
| Degradation Products | Degradation products such as deamidation can form during exposure to high temperatures or pH excursions. Use of backbone protecting groups minimizes the formation of degradation products during the SPPS build phase. | Deamidation impurities may have partial removal through drug substance chromatography purification. Established PARs ensure that deamidation impurity levels can be reduced to the extent that specification limits in the drug substance are met. |

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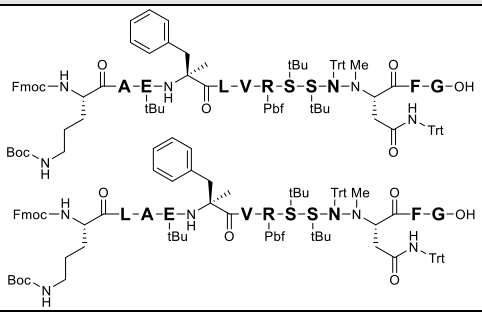
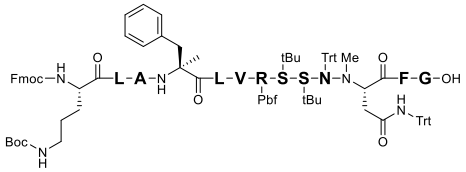
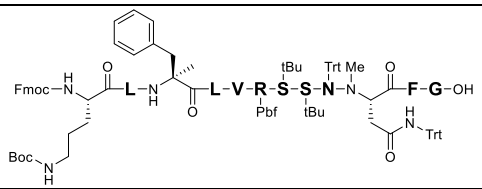
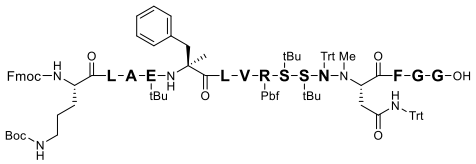
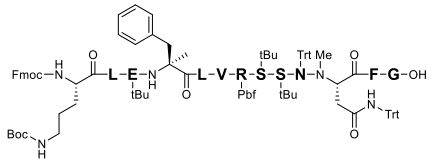
**Table 7.2:** Select Impurities

| RRT | Impurity | Structure | Source | Point of Control |
|---|---|---|---|---|
| 0.27 | L2 DIC-22-24 *(mass 718.4 or -2103.1 Δ)* |  | Process | • NMeAsn22 activation and coupling process controls.<br>• Fragment L2 unspecified impurity. |
| 0.27 | L2 Trt cation *(mass 242.1 or -2579.3 Δ)* | Unknown | N/A | • Fragment L2 unspecified impurity. |
| 0.47 | L2 Ac-21-24 *(mass 990.4 or -1831.0 Δ)* |  | Process | • Acetic acid amount in IPAC wash solvent.<br>• SPPS process controls.<br>• Fragment L2 unspecified impurity. |
| 0.51 | L2 M-805 Da *(mass 2016.0 or -805.4 Δ)* | Unknown | N/A | • Fragment L2 unspecified impurity. |
| 0.52 | L2 M-1053 Da *(mass 1768.9 or -1052.5 Δ)* | Unknown | N/A | • Fragment L2 unspecified impurity. |
| 0.53 | L2 DIC-17-24 *(mass 1868.0 or -953.5 Δ)* |  | Process | • Val17 activation and coupling process controls.<br>• Fragment L2 unspecified impurity. |
| 0.58 | L2 M-436 Da *(mass 2385.3 or -436.2 Δ)* | Unknown | N/A | • Fragment L2 unspecified impurity. |
| 0.59 | L2 DIC-14-24 *(mass 2327.3 or -494.2 Δ)* |  | Process | • Glu14 activation and coupling process controls.<br>• Fragment L2 unspecified impurity. |
| 0.59 | L2 H-11-24 *(mass 2599.4 or -222.1 Δ)* |  | Process Degradation | • SPPS and soft cleavage process controls.<br>• Fragment L2 unspecified impurity.<br>• Fragment L2 storage conditions. |
| 0.60 | L2 NMeAsp22 *(mass 2580.3 or -241.2 Δ)* |  | Process | • Coupling and Fmoc deprotection process controls to limit the conversion of NMeAsn22 to aspartic acid through the aspartimide pathway.<br>• Fragment L2 unspecified impurity. |
| 0.63 | L2 M-694 Da *(mass 2127.1 or -694.4 Δ)* | Unknown; mass aligns with a piperidine adduct at αMePhe15. | N/A | • Fragment L2 unspecified impurity. |
| 0.66 | L2 M-894 Da *(mass 1927.0 or -894.5 Δ)* | Unknown | N/A | • Fragment L2 unspecified impurity. |
| 0.66 | L2 Orn11(des-Boc) *(mass 2721.4 or -100.0 Δ)* |  | Starting Material Process Degradation | • Des-Boc unspecified impurity in starting material.<br>• Soft cleavage process controls.<br>• Fragment L2 unspecified impurity.<br>• Fragment L2 storage conditions. |

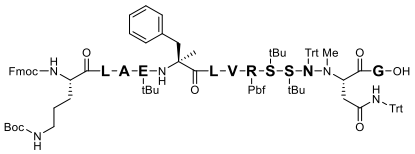
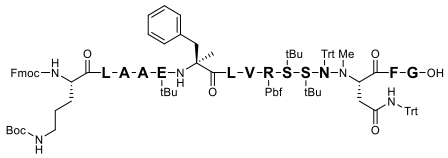
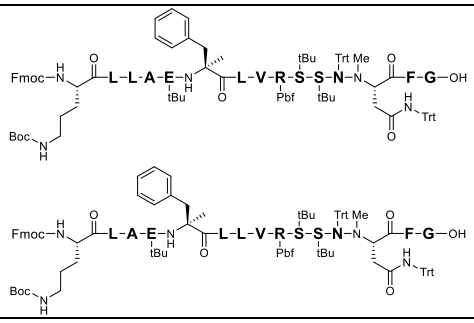
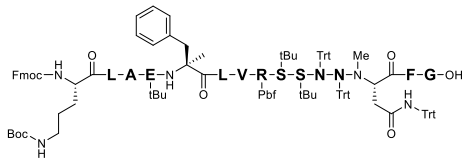
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| RRT | Impurity | Structure | Source | Point of Control |
|---|---|---|---|---|
| 0.67 | L2 Ac-15-24 *(mass 2058.0 or -763.4 Δ)* |  | Raw Material Process | • Acetic acid amount in IPAC wash solvent.<br>• SPPS process controls.<br>• Fragment L2 unspecified impurity. |
| 0.67 | L2 M-855 Da *(mass 1966.0 or -855.4 Δ)* | Unknown; mass aligns with a piperidine adduct at Leu16. | N/A | • Fragment L2 unspecified impurity. |
| 0.69 | L2 M-779 Da *(mass 2042.0 or -779.4 Δ)* | Unknown; mass aligns with an imine adduct at αMePhe15. | N/A | • Fragment L2 unspecified impurity. |
| 0.69 | L2 M-666 Da *(mass 2155.1 or -666.4 Δ)* | Unknown | N/A | • Fragment L2 unspecified impurity. |
| 0.69 | L2 M-521 Da *(mass 2300.2 or -521.3 Δ)* | Unknown | N/A | • Fragment L2 unspecified impurity. |
| 0.71 | L2 Ac-13-24 *(mass 2314.2 or -507.2 Δ)* |  | Raw Material Process | • Acetic acid amount in IPAC wash solvent.<br>• SPPS process controls.<br>• Fragment L2 unspecified impurity. |
| 0.72 | L2 M-567 Da *(mass 2254.1 or -567.4 Δ)* | Unknown; mass aligns with an oxidized Fmoc at αMePhe15. | N/A | • Fragment L2 unspecified impurity. |
| 0.75 | L2 Ac-14-24 *(mass 2243.2 or -578.3 Δ)* |  | Raw Material Process | • Acetic acid amount in IPAC wash solvent.<br>• SPPS process controls.<br>• Fragment L2 unspecified impurity. |
| 0.78 | L2 M-410 Da *(mass 2411.2 or -410.2 Δ)* | Unknown; mass aligns with an imine adduct at Leu12. | N/A | • Fragment L2 unspecified impurity. |
| 0.78 | L2 des-(Asn21-NMeAsn22) *(mass 2095.1 or -726.3 Δ)* |  | Process | • Asn21 coupling and Fmoc deprotection process controls to prevent impurity formation via the diketopiperazine (DKP) mechanism.<br>• Fragment L2 specified impurity. |
| 0.79 | L2 des-Asn21 *(mass 2465.3 or -356.1 Δ)* |  | Process | • NMeAsn22 Fmoc deprotection and Asn21 coupling process controls.<br>• In-process ninhydrin test to confirm coupling.<br>• Fragment L2 unspecified impurity. |
| 0.79 | L2 M-382 Da *(mass 2439.2 or -382.2 Δ)* | Unknown; mass aligns with an oxidized Fmoc at Glu14. | N/A | • Fragment L2 unspecified impurity. |
| 0.87 | L2 des-αMePhe15 *(mass 2660.4 or -161.1 Δ)* |  | Process | • Leu16 Fmoc deprotection and αMePhe15 coupling process controls.<br>• In-process ninhydrin test to confirm coupling.<br>• Fragment L2 unspecified impurity. |

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| RRT | Impurity | Structure | Source | Point of Control |
|---|---|---|---|---|
| 0.87<br>0.89 | L2 des-Leu12/16<br>*(mass 2708.4 or -113.1 Δ)* |  | Process | • Ala13 or Val17 Fmoc deprotection and Leu12 or Leu16 coupling process controls.<br>• In-process ninhydrin test to confirm coupling.<br>• Fragment L2 unspecified impurity. |
| 0.89 | L2 M-369 Da<br>*(mass 2452.3 or -369.2 Δ)* | Unknown | N/A | • Fragment L2 unspecified impurity. |
| 0.89 | L2 des-Glu14<br>*(mass 2636.4 or -185.1 Δ)* |  | Process | • αMePhe15 Fmoc deprotection and Glu14 coupling process controls.<br>• In-process ninhydrin test to confirm coupling.<br>• Fragment L2 unspecified impurity. |
| 0.89 | L2 des-(Ala13-Glu14)<br>*(mass 2565.4 or -256.1 Δ)* |  | Process | • Ala13 coupling and Fmoc deprotection process controls to prevent impurity formation via the diketopiperazine (DKP) mechanism.<br>• Fragment L2 unspecified impurity. |
| 0.92 | L2 M+16 Da<br>*(mass 2837.5 or +16.0Δ)* | Unknown; potentially and oxidized Fmoc group at Orn11. | Process | • Fragment L2 unspecified impurity. |
| 0.94 | L2 M+914 Da<br>*(mass 3735.9 or +914.4 Δ)* | Unknown | N/A | • Fragment L2 unspecified impurity. |
| 0.94<br>0.96<br>0.98<br>1.02<br>1.02<br>1.06 | L2 Isomer<br>*(mass 2821.5)* | Multiple Isomers | Starting Material Process | • D-isomer specified impurity in starting materials.<br>• Fmoc deprotection and activation/coupling process controls.<br>• Fragment L2 unspecified impurity. |
| 0.96 | L2 endo-Gly24<br>*(mass 2878.5 or +57.0 Δ)* |  | Starting Material Process | • H-Gly-OH specified impurity in starting material.<br>• Gly24 loading process controls.<br>• Fragment L2 specified impurity. |
| 0.96 | L2 endo-β-Ala13<br>*(mass 2892.5 or +71.0 Δ)* | OBJ The β-alanine could also be present to the N-terminus of the alanine residue | Starting Material | • H- β-alanine specified impurity in starting materials.<br>• Fragment L2 unspecified impurity. |
| 0.96 | L2 des-Ala13<br>*(mass 2750.4 or -71.0 Δ)* |  | Process | • Glu14 Fmoc deprotection and Ala13 coupling process controls.<br>• In-process ninhydrin test to confirm coupling.<br>• Fragment L2 unspecified impurity. |

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| RRT | Impurity | Structure | Source | Point of Control |
|---|---|---|---|---|
| 0.97 | L2 des-Phe23 *(mass 2674.4 or -147.1 Δ)* |  | Process | • Gly24 Fmoc deprotection and Phe23 coupling process controls.<br>• In-process ninhydrin test to confirm coupling.<br>• Fragment L2 unspecified impurity. |
| 0.97 | L2 M-14 Da *(mass 2807.4 or -14.0 Δ)* | Unknown | N/A | • Fragment L2 unspecified impurity. |
| 1.02 | L2 M-101 Da *(mass 2720.4 or -101.0 Δ)* | Unknown | N/A | • Fragment L2 unspecified impurity. |
| 1.03 | L2 endo-Ala13 *(mass 2892.5 or +71.0 Δ)* |  | Starting Material Process | • H-Ala-OH specified impurity in starting materials.<br>• Glu14 Fmoc deprotection wash and Ala13 coupling process controls.<br>• Fragment L2 unspecified impurity. |
| 1.03 | L2 M+14 Da *(mass 2835.5 or +14.0 Δ)* | Unknown | N/A | • Fragment L2 unspecified impurity. |
| 1.05 | L2 M-14 Da *(mass 2807.4 or -14.0 Δ)* | Unknown | N/A | • Fragment L2 unspecified impurity. |
| 1.06 | L2 M+90 Da *(mass 2911.5 or +90.1 Δ)* | Unknown; mass aligns with an addition of both Phe23 and Gly24. | N/A | • Fragment L2 unspecified impurity. |
| 1.08<br>1.14 | L2 endo-Leu *(mass 2934.5 or +113.1 Δ)* |  | Starting Material Process | • H-Leu-OH specified impurity in starting material.<br>• Ala13/Val17 Fmoc deprotection wash and Leu12/Leu16 coupling process controls.<br>• Fragment L2 unspecified impurity. |
| 1.09 | L2 M+56 Da *(mass 2877.5 or +56.1 Δ)* | N/A | Process | • Soft cleavage process controls<br>• Fragment L2 unspecified impurity. |
| 1.10 | L2 M+574 Da *(mass 3395.7 or +574.3 Δ)* | Unknown; mass aligns with an addition of NMeAsn22, Phe23, and Gly24. | N/A | • Fragment L2 unspecified impurity. |
| 1.11 | L2 M+185 Da *(mass 3006.6 or +185.1 Δ)* | Unknown | N/A | • Fragment L2 unspecified impurity. |
| 1.17 | L2 endo-Asn21 *(mass 3177.7 or +356.2 Δ)* |  | Starting Material Process | • H-Asn(Trt)-OH specified impurity in starting material.<br>• NMeAsn22 Fmoc deprotection and Asn21 coupling process controls.<br>• Fragment L2 unspecified impurity. |

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# 8. Other Historical Knowledge & Critical Learning

Additional process understanding may be generated through continued process verification activities, commercial manufacturing experience and future characterization studies. Resulting knowledge may be used to refine PARs, reassess CPP classifications and enhance process robustness throughout the product lifecycle.

# 9. Process Hazards

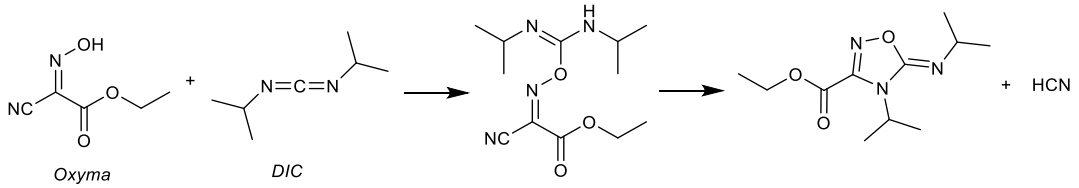
Refer to the Safety Data Sheet for information on Eloralintide Fragment L2 intermediate.

## 9.1 Occupational Exposure Limit (OEL)

The occupational exposure limit (OEL) for Fragment L2 is not established; however, preliminary guidance is to control exposures to between 10 and 100 mg/m$^3$ until an OEL is established.

## 9.2 Hazardous Compounds

**Table 9.2:** Hazardous Compounds

| Compound | Information |
|---|---|
| Hydroxybenzotriazole (HOBt) | HOBt is generated from PyBOP during the coupling of Fmoc-Phe-OH. HOBt is an explosive in its anhydrous form. See Figure 3.2.5. |
| Hydrogen Cyanide (HCN) | HCN may be generated from Oxyma and DIC under ambient conditions during the activation and coupling steps as a side-reaction.[2] HCN is a health hazard.   |

[2] Generation of Hydrogen Cyanide from the Reaction of Oxyma (Ethyl Cyano(hydroxyimino)acetate) and DIC (Diisopropylcarbodiimide); A. McFarland et al., *Organic Process Research & Development 2019 23 (9), 2099-2105*.

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Document Approval Signatures

Approved Date: 10 Nov 2026

| | |
|---|---|
| Approval<br>Verdict: Approve | Luke O'Sullivan<br>(L035019@lilly.com)<br>Approver<br>10-Nov-2026 08:45:56 GMT+0000 |
| Approval<br>Verdict: Approve | Chloe Coveney<br>(L106512@lilly.com)<br>Approver<br>10-Nov-2026 16:06:18 GMT+0000 |
| Approval<br>Verdict: Approve | Michelle Cronin-Carroll<br>(C203669@lilly.com)<br>Approver<br>10-Nov-2026 16:16:29 GMT+0000 |
| Approval<br>Verdict: Approve | Anita Cremin<br>(L046349@lilly.com)<br>Quality Approver<br>10-Nov-2026 16:32:18 GMT+0000 |
| Reviewer / Approver Additional Details | Luke O'Sullivan is signing as Data Integrity Reviewer (DIR).<br><br>Michelle Cronin-Carroll, Chloe Coveney & Anita Cremin are signing as Lilly Approvers. |

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